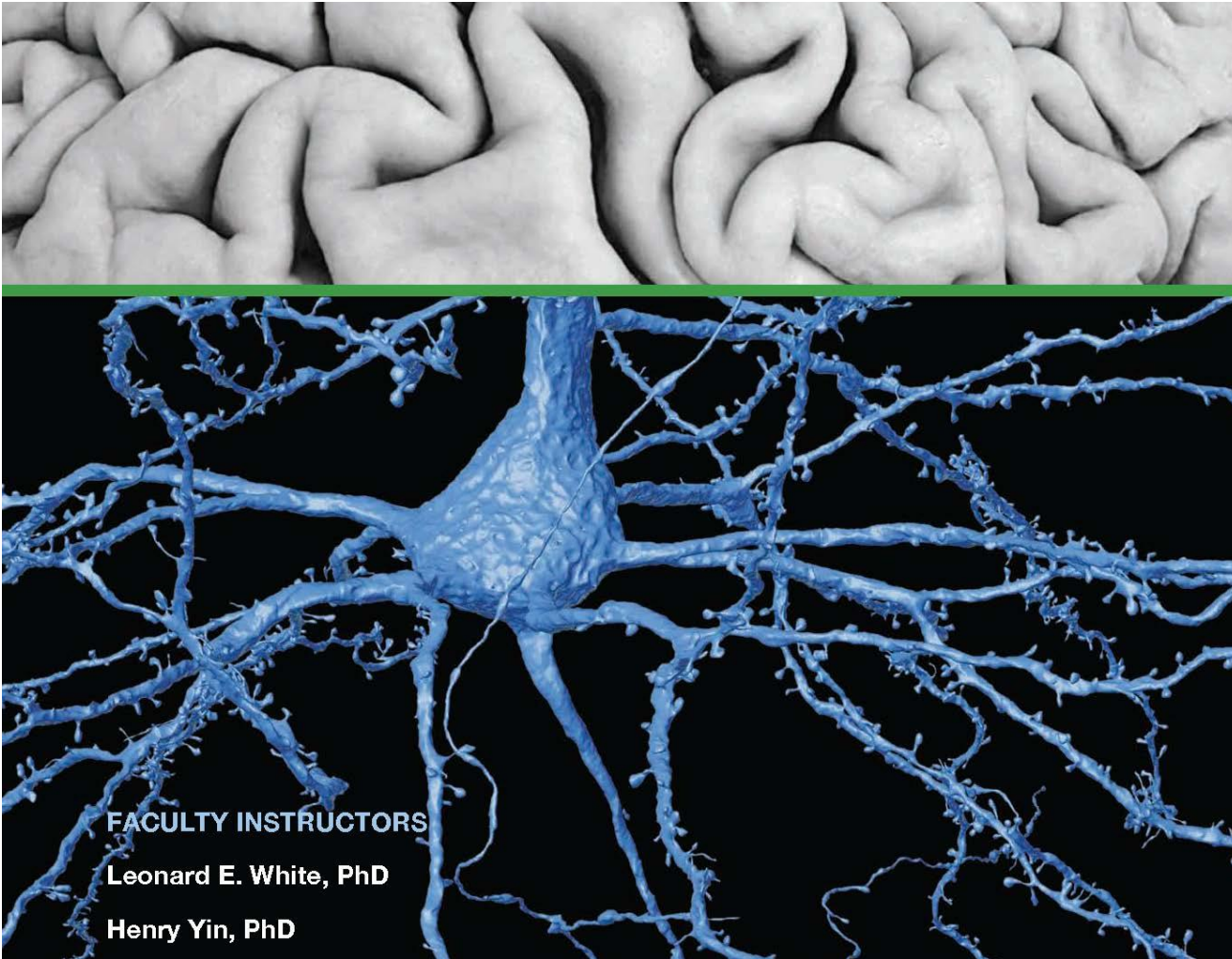


Welcome to NEUROSCI 206L

Introduction to Systems Neuroscience



Sit wherever you like.

*Please complete the
“Getting to Know
You” survey in Canvas*

(if you have not already done so)

Welcome to NEUROSCI 206L

Introduction to Systems Neuroscience

- instructional faculty and teaching assistants
- what is “Systems Neuroscience”?
- why did we create this new course?
- course format, resources, logistics, and policies (syllabus)
- team formation
- getting ready for Week 1 in 206L
- getting help

NEUROSCI 206L Introduction to Systems Neuroscience

Faculty Instructors

- **Leonard E. White, PhD**
- **Henry Yin, PhD**

Assisting Instructors

- **Alev Brigande (doctoral student)**
- **Hanchang Zheng (doctoral student)**
- **Roderick Bailey (Neuroscience Teaching Lab manager)**

What is “Systems Neuroscience”?

The domain within the broader fields of neuroscience and biology that studies the **structure and function of neural circuits**, which interact at various levels of the nervous system to produce behavior.

- neural circuits are typically organized within **systems of neural connections** that process the same modalities of information (e.g., vision, touch, taste) or integrate across modalities (e.g., motor control, motivation, attention)
- considers components of the peripheral nervous system (nerves, ganglia, associated effectors) and the central nervous system (brain, spinal cord, retina)
- such systems of neural connections are often organized along **sensory and motor pathways** that convey neural signals between peripheral parts of the body and relevant processing circuits in the brain
- typically involves anatomical, physiological, behavioral, computational, and theoretical studies of brain and behavior using a variety of methodologies

Why did we create this course?

- Our programmatic assessment work in recent years consistently revealed several relevant findings from graduating Neuroscience majors when asked to reflect on our foundational curriculum:
 - there was insufficient coverage of **sensory and motor systems**, and computational neuroscience
 - there was insufficient exposure to **human neuroanatomy**
 - there was too much redundancy between our “gateway” course and the pre-existing two foundational courses (NEUROSCI 217D and NEUROSCI 223)
- NEUROSCI 206L was created to fulfill these goals:
 - provide a systematic overview of the **organization and function of sensory and motor systems**
 - provide systems-based explorations of **human neuroanatomy using real human specimens**
 - **conduct physiological experiments** demonstrating mechanisms of sensation and action
 - ensure that **all Neuroscience majors** will have these learning experiences in the newly reformed foundational curriculum for the major

Scope of the course

- < fall break: human neuroanatomy and sensory systems
 - Introducing systems neuroscience and getting to know the human brain
 - General principles of sensory system organization
 - Visual system
 - Auditory system
 - Vestibular system
 - Somatic sensory systems
 - Chemical senses: olfaction and taste

Scope of the course

- > fall break: human neuroanatomy and motor control
 - midterm (first day back from break!)
 - Lower motor control of movement
 - Upper motor control of movement
 - Basal ganglia
 - Cerebellum
 - Eye movements
 - Visceral motor system (autonomic nervous system)
 - Comprehensive final exam (Dec. 13th)

NEUROSCI 206L Introduction to Systems Neuroscience

Weekly format

Mondays

4:40-5:55

Overview lecture

- faculty will layout content to be assimilated for the week's activities
- attendance optional
- no Zoom recording
- lectures slides or handouts will be posted in advance of class time

Wednesdays

4:40-5:55

Team-Based Learning

- Readiness Assurance (individual/team phases)
- based on overview lecture and assigned readings
- debrief and Q&A
- direction for the week's laboratory activities

Thursdays or Fridays

1:25-2:40

8:30-9:45

Laboratory activities

- explorations of human brain specimens and digital data repositories
- physiological experiments on teammate volunteers (Nerve conduction, EMG, EOG, sensory testing, sensorimotor plasticity)

NEUROSCI 206L Introduction to Systems Neuroscience

Team-Based Learning

Monday

Learner Preparation (knowledge is acquired through lectures, assigned readings, and lab materials)

Wednesday

Readiness Assurance (knowledge is assessed individually, then in teams with immediate feedback)

Q&A (faculty expertise consolidates learning and ensures readiness for knowledge application)

Thursday/Friday

Team Application (Lab) (knowledge is applied through brain labs and physiological experiments)



NEUROSCI 206L Introduction to Systems Neuroscience

Course resources

- Canvas course site
- *Neuroscience, 7th Edition* (Oxford University Press) / Oxford Insight
 - hardcopy: \$129-200 on Amazon
 - ebook: \$85 for 6 months or \$103 for “lifetime”
 - Oxford Insight (digital book, *Sylvius*, animations, practice quizzes): \$85 for 6 months
 - earlier editions (4th – 6th) are adequate (used from \$15 on Amazon)
- *Sylvius* (digital atlas of the human CNS): FREE for Duke
- Lt (course software for physiological laboratories): FREE for Duke

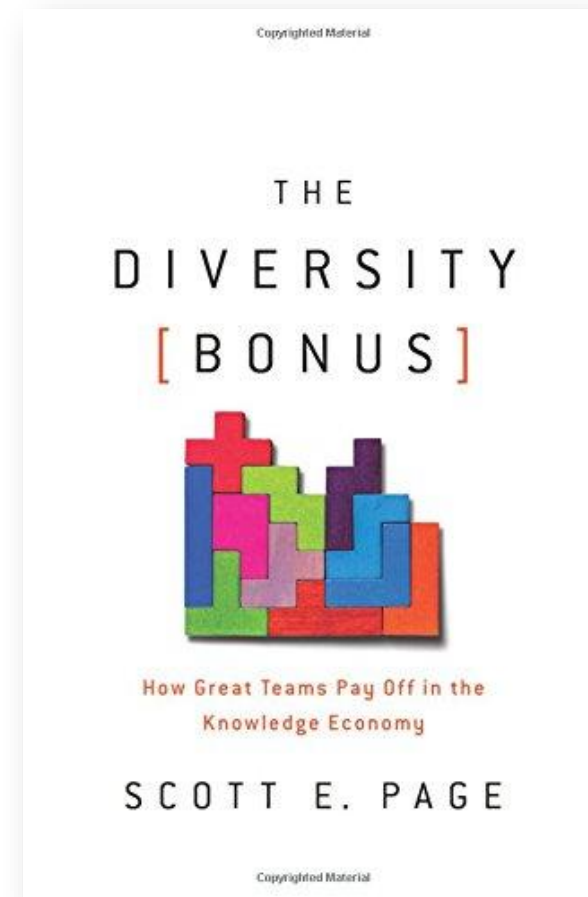
NEUROSCI 206L Introduction to Systems Neuroscience

Course policies

- Attendance
 - lecture: optional (we will not make recordings)
 - TBL activities: assessments require attendance (iRA = tRA for excused absences)
 - laboratory activities: hands-on learning requires attendance
- Lab attire
 - per Duke Lab Safety policy, closed-toed shoes are required
- Artificial intelligence
 - prohibited for all in-class assessments (RA-RA, midterm, final)
 - may be used on Lab Reports, with declaration of AI methods (see syllabus)

Team formation

- Primary principal: maximize
cognitive diversity



Getting ready for Week 1

- work through the “Week 0” module on the Canvas
 - gather your resources (please read the syllabus)
 - familiarize yourself with these resources
- Study the foundational lessons
 - review and study Monday’s lecture content
 - read and study assigned readings
- Preview the Lab Protocol
 - expect Wednesday’s Readiness Assurance to cover lab protocol
 - review guidelines for the week’s Lab Report

NEUROSCI 206L Introduction to Systems Neuroscience

Getting help

- ***Before class***

- post questions to Discussion forum on Canvas course site
- come early to class and bring questions and/or conversation

- ***In class***

- Mondays: be attentive; be curious; ask questions
- Wednesdays: embrace the journey; get the most out of your teamwork
- Labs: work strategically and collaboratively; ask questions

- ***Outside of class***

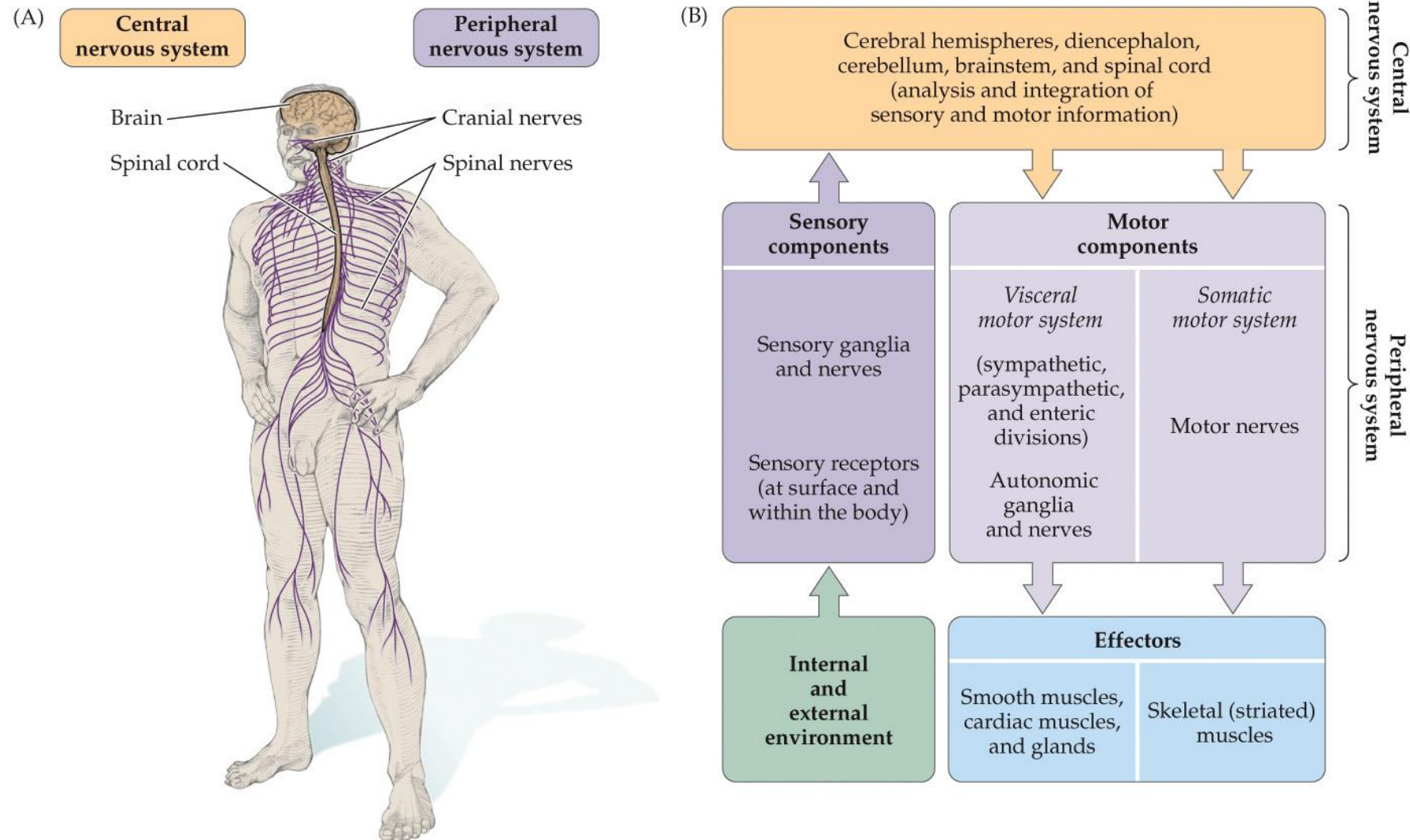
- regular help/review sessions with assisting instructors
- group and individual meetings with faculty may be scheduled

Getting to Know the Human Brain

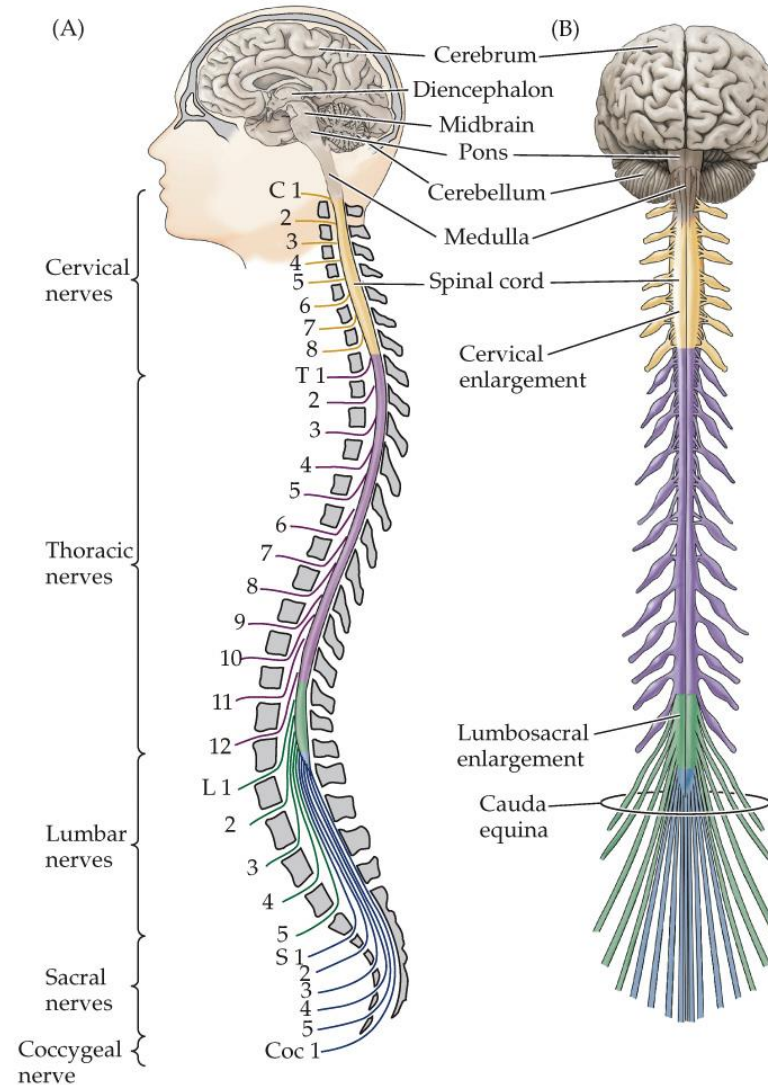
Learning objectives

- Identify the basic, embryological subdivisions of the human brain, including the brainstem and the ventricular spaces that are visible in the mid-sagittal plane.
- Localize the surface features that define the boundaries between the contiguous lobes of the cerebral hemisphere.
- Localize and characterize the location and morphology of the principal gyri and sulci on the lateral, medial, and ventral aspects of the cerebral hemispheres.

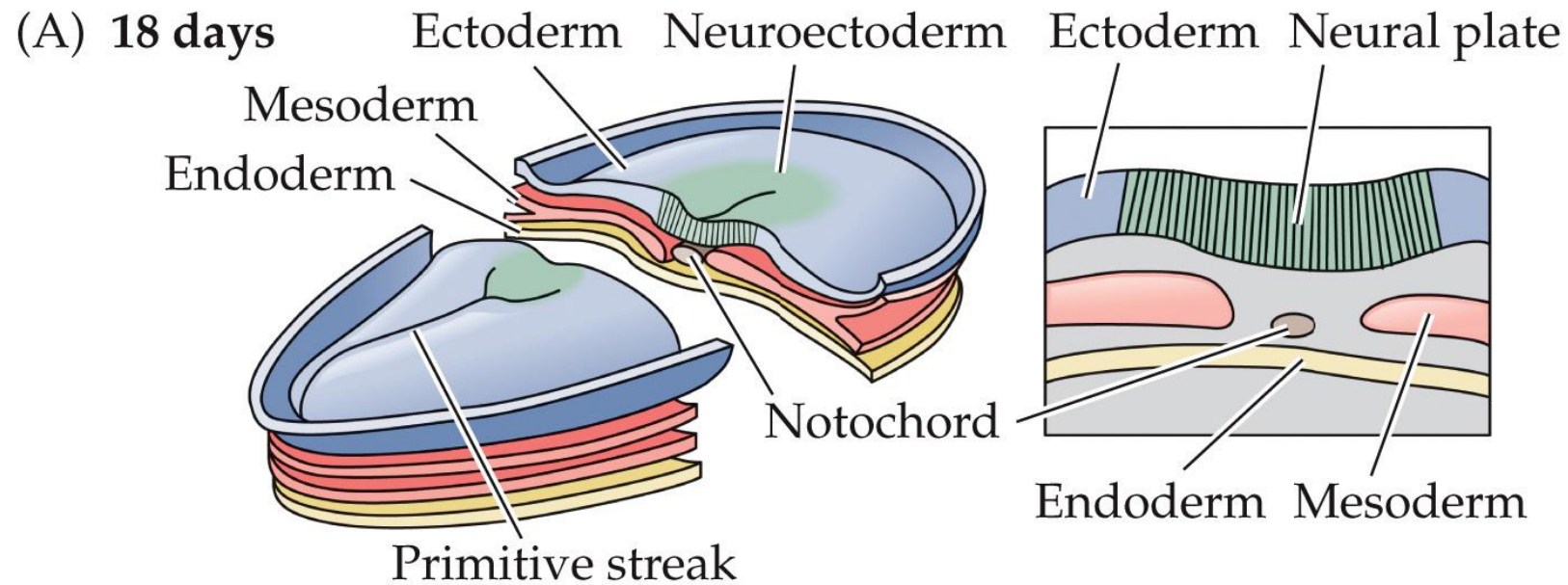
Basic divisions of the human nervous system



Basic divisions of the human nervous system

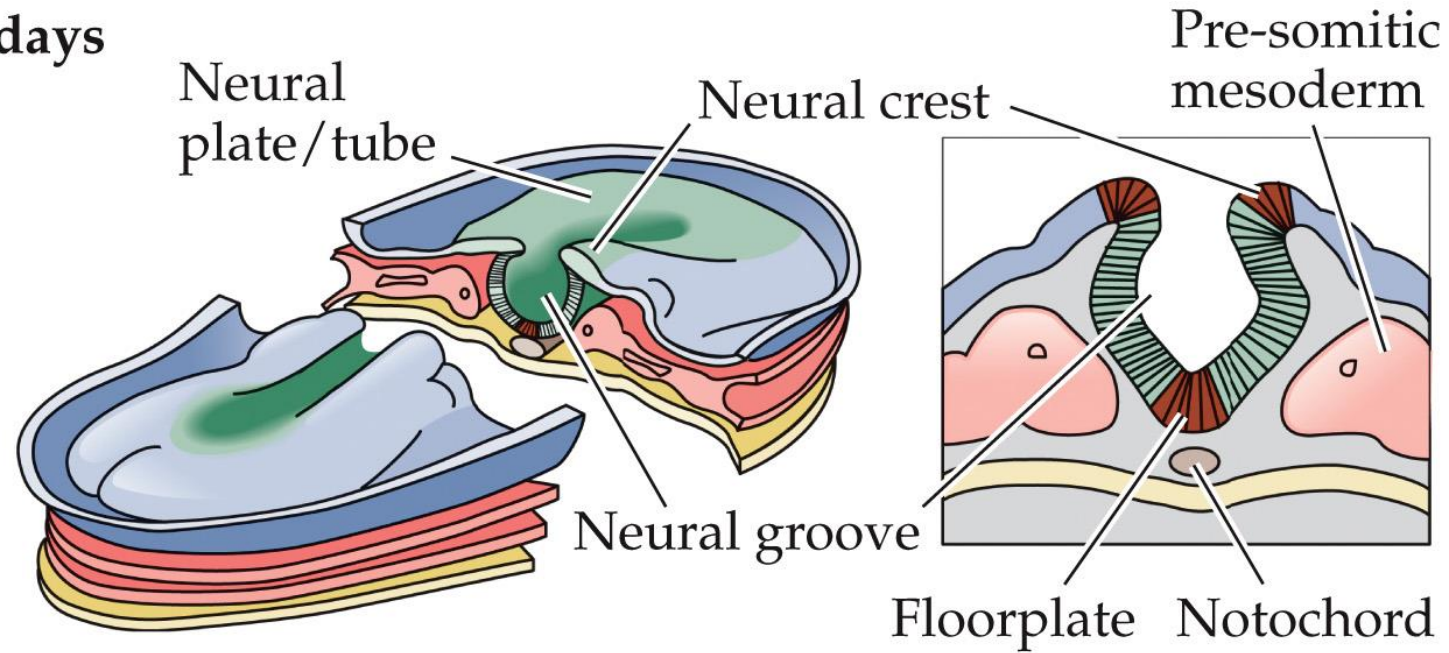


Formation of the human nervous system



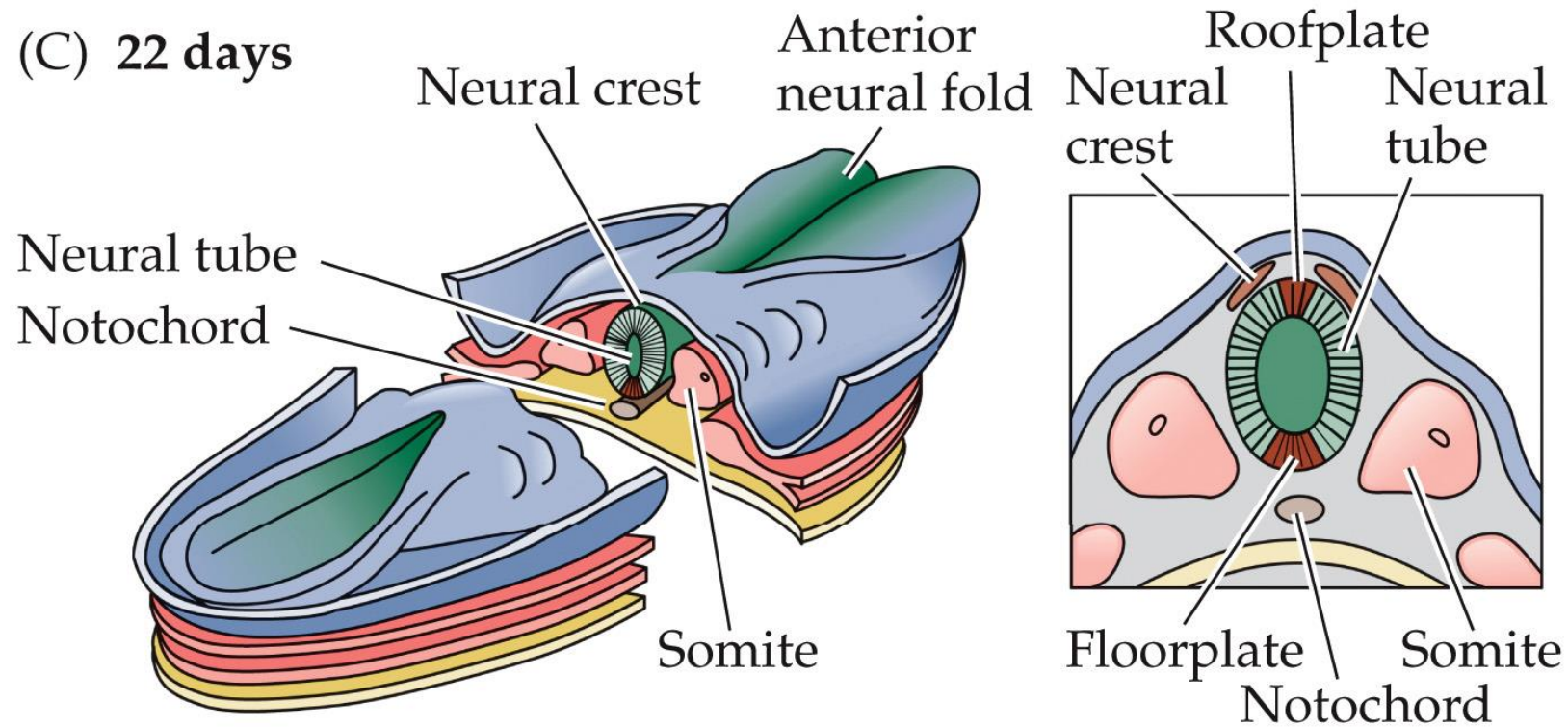
Formation of the human nervous system

(B) 20 days

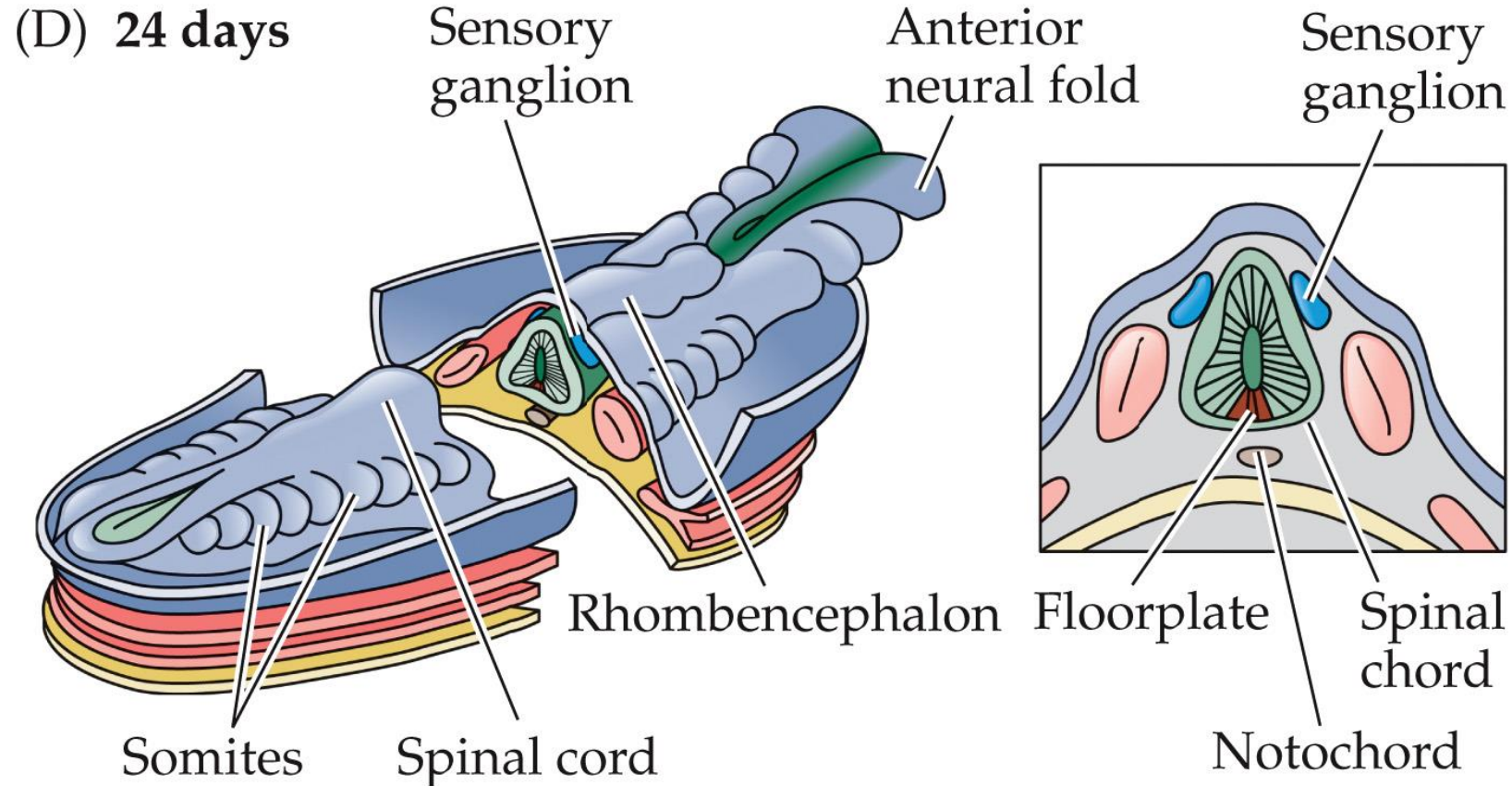


NEUROSCIENCE 6e, Figure 22.1 (Part 2)
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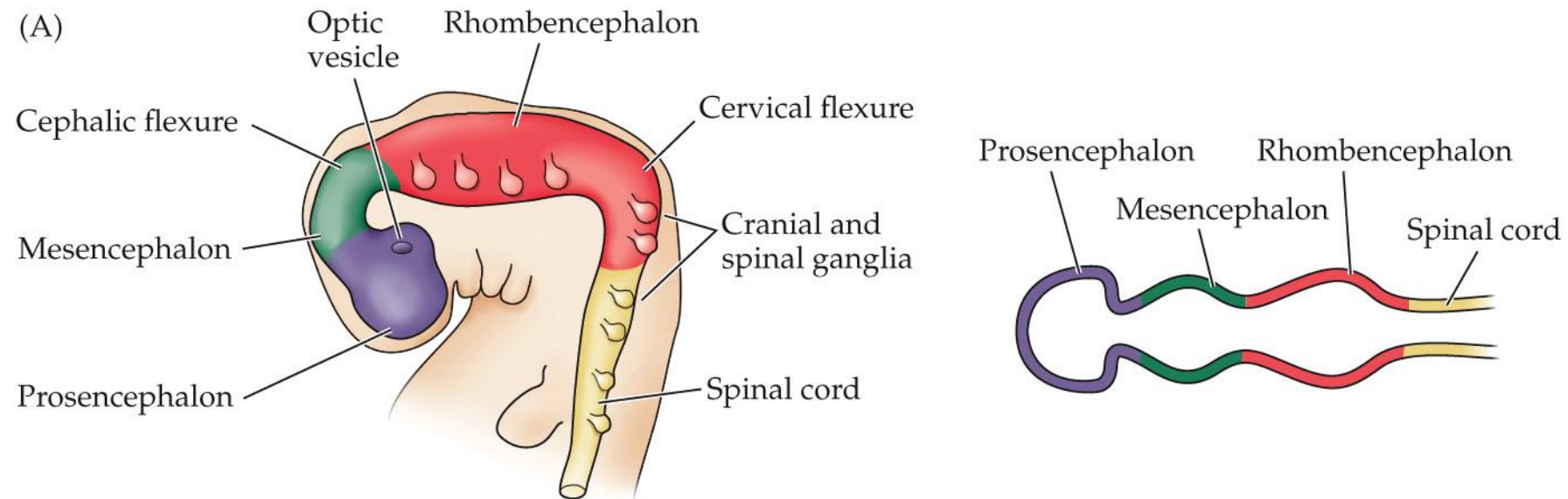
Formation of the human nervous system



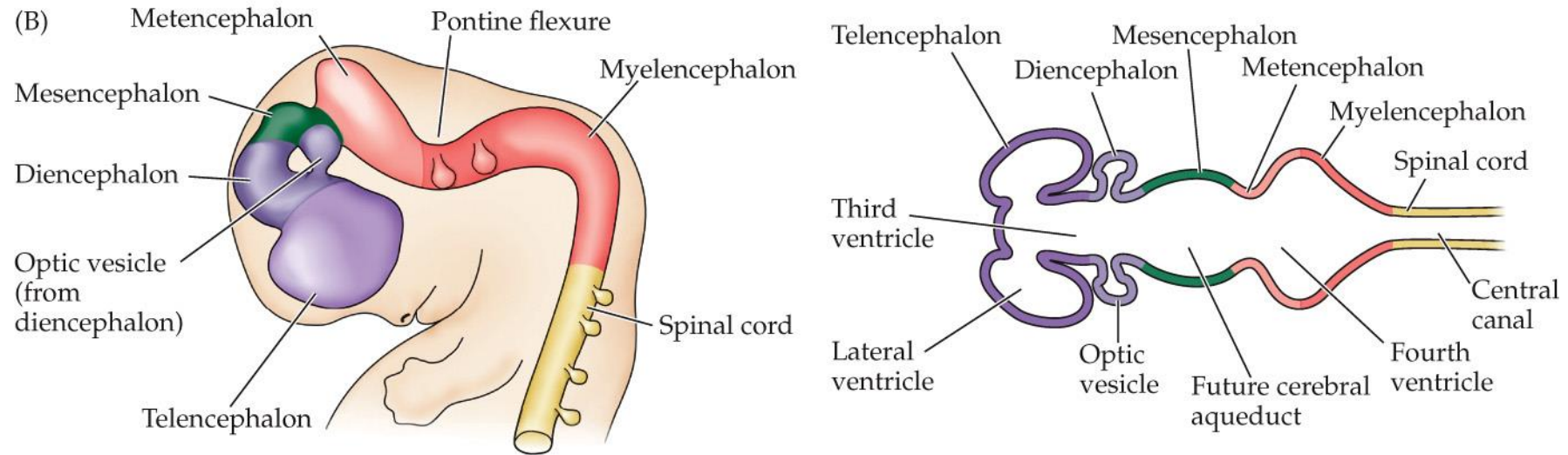
Formation of the human nervous system



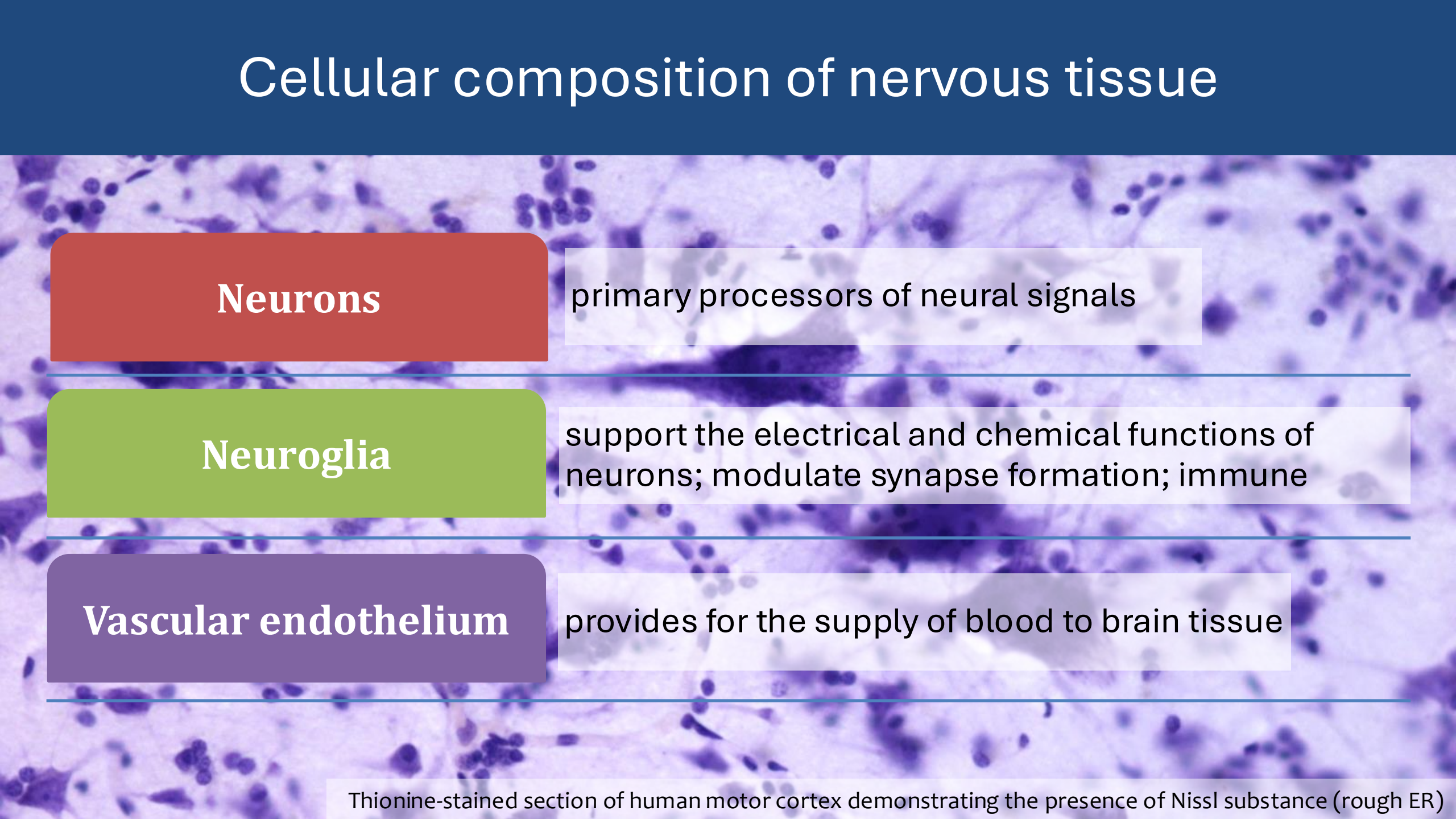
Formation of the human nervous system



Formation of the human nervous system



Cellular composition of nervous tissue



Neurons

primary processors of neural signals

Neuroglia

support the electrical and chemical functions of neurons; modulate synapse formation; immune

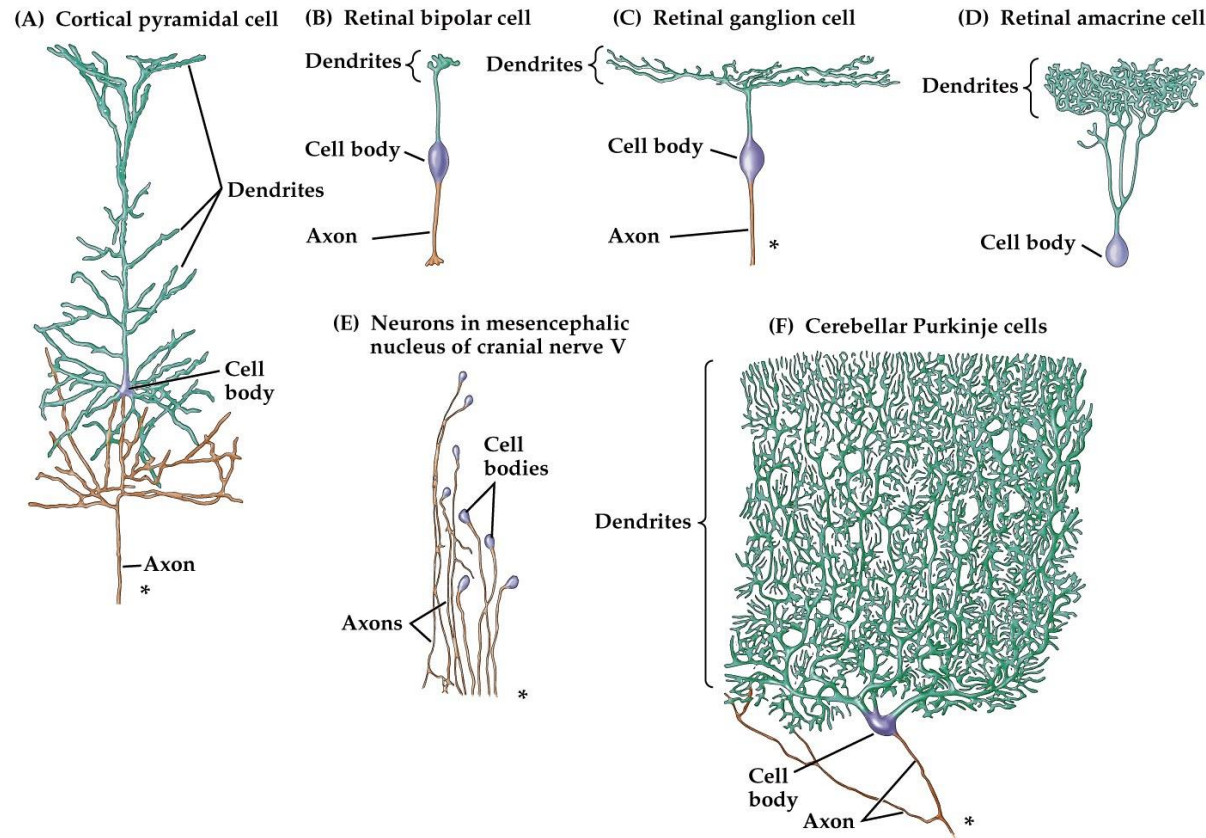
Vascular endothelium

provides for the supply of blood to brain tissue

Thionine-stained section of human motor cortex demonstrating the presence of Nissl substance (rough ER)

Neurons

Fundamental unit of function in the CNS

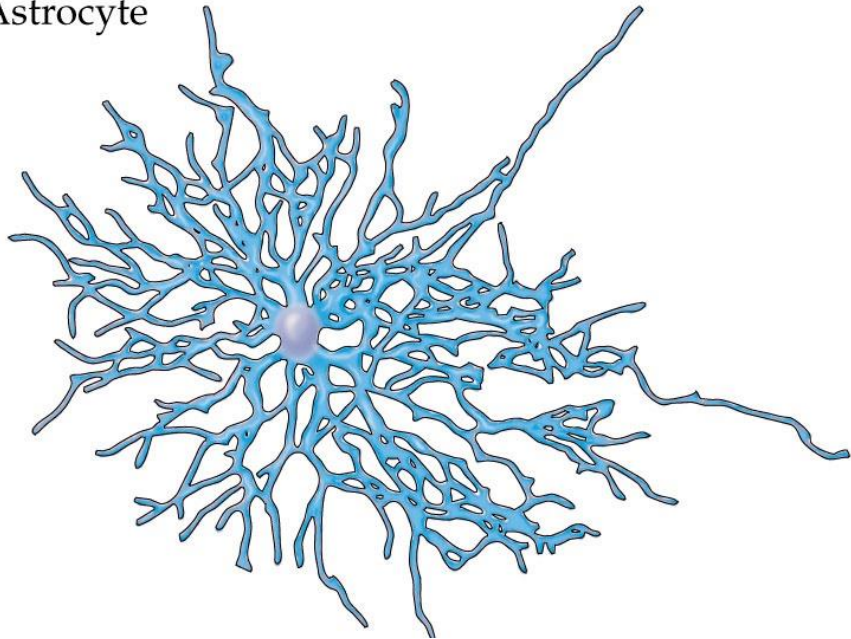


- maintain all the metabolic machinery common to other cells of the body
- but, distinct from most other cells:
 - rich diversity in morphology
 - “excitable” with bioelectrical properties
 - specializations for intercellular communication

Neuroglia (“Glia”)

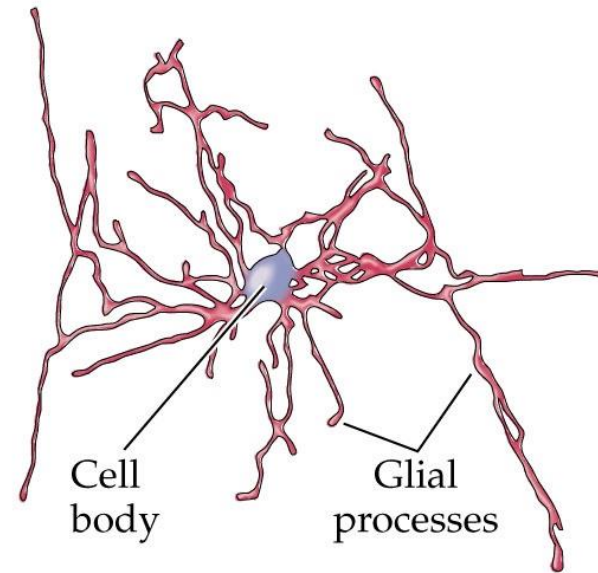
Support and modulate the functions of neurons

(A) Astrocyte



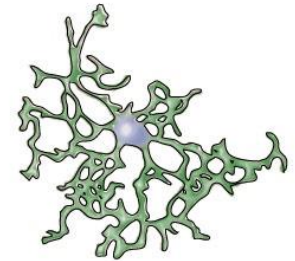
- support the chemical functions of neurons
- modulate synapse formation
- participate in blood-brain barrier
- participate in scar formation

(B) Oligodendrocyte



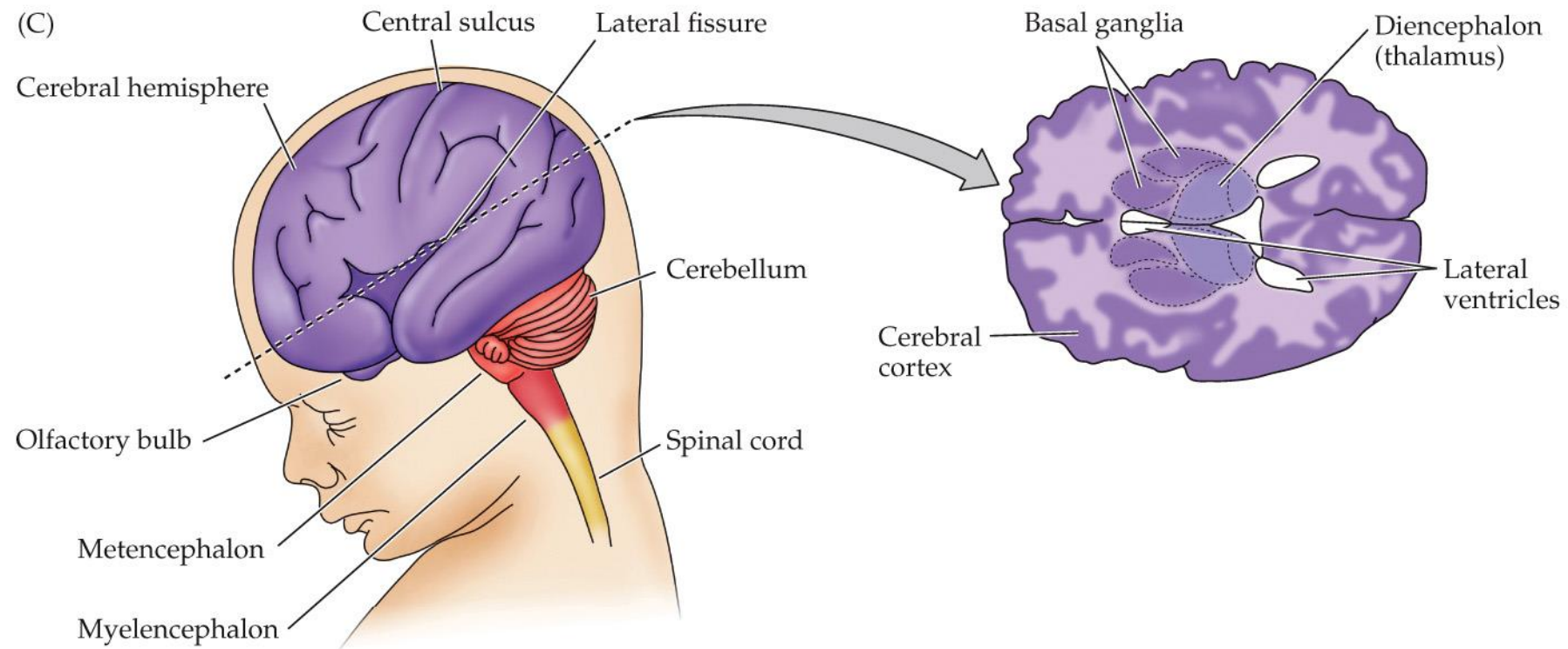
- make myelin (insulation), facilitating the rapid propagation of action potentials
- shape axon regeneration

(C) Microglial cell

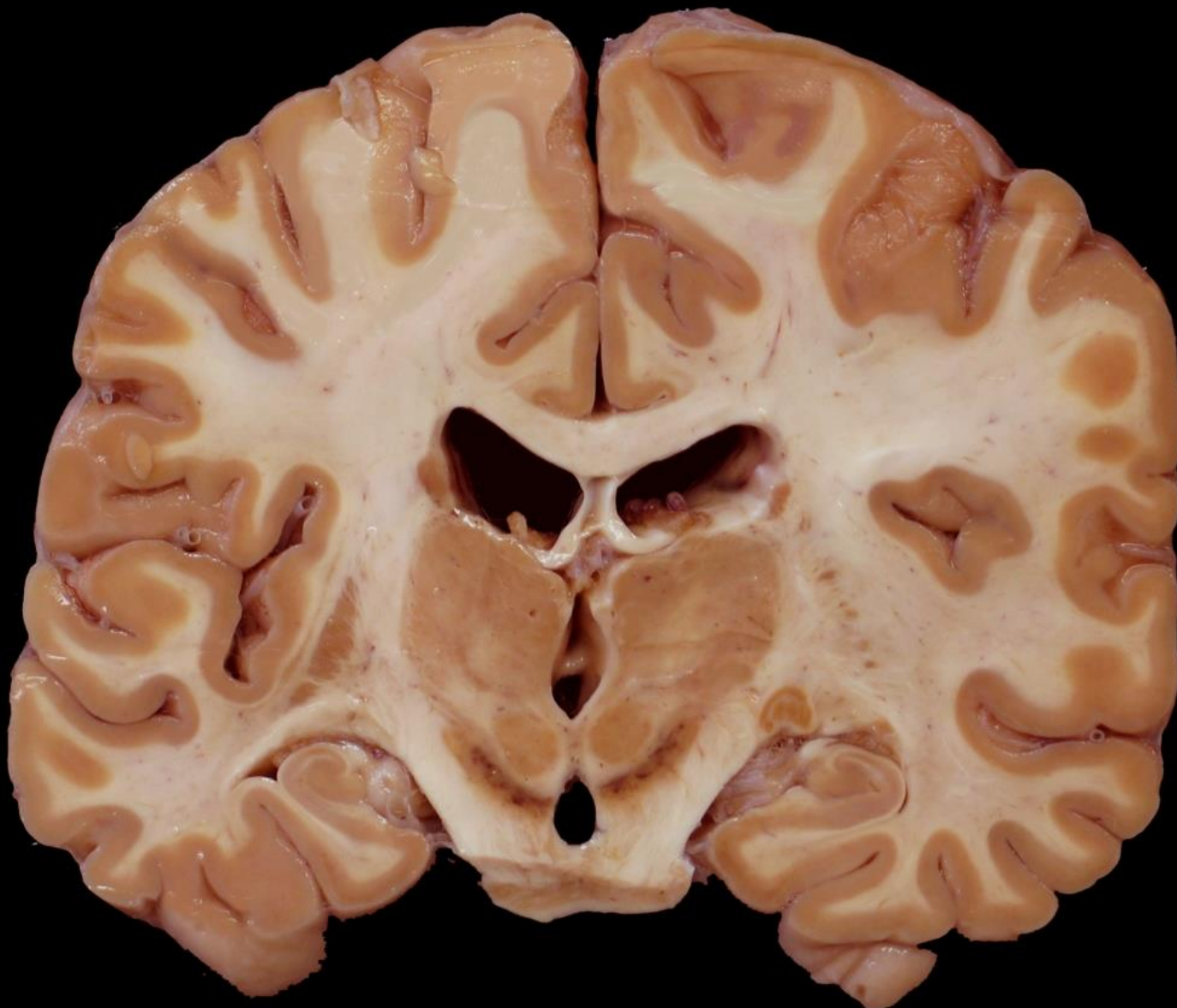


- resident immune cell of the CNS
- inflammation
- wound healing

Formation of the human nervous system



Gray/White Matter

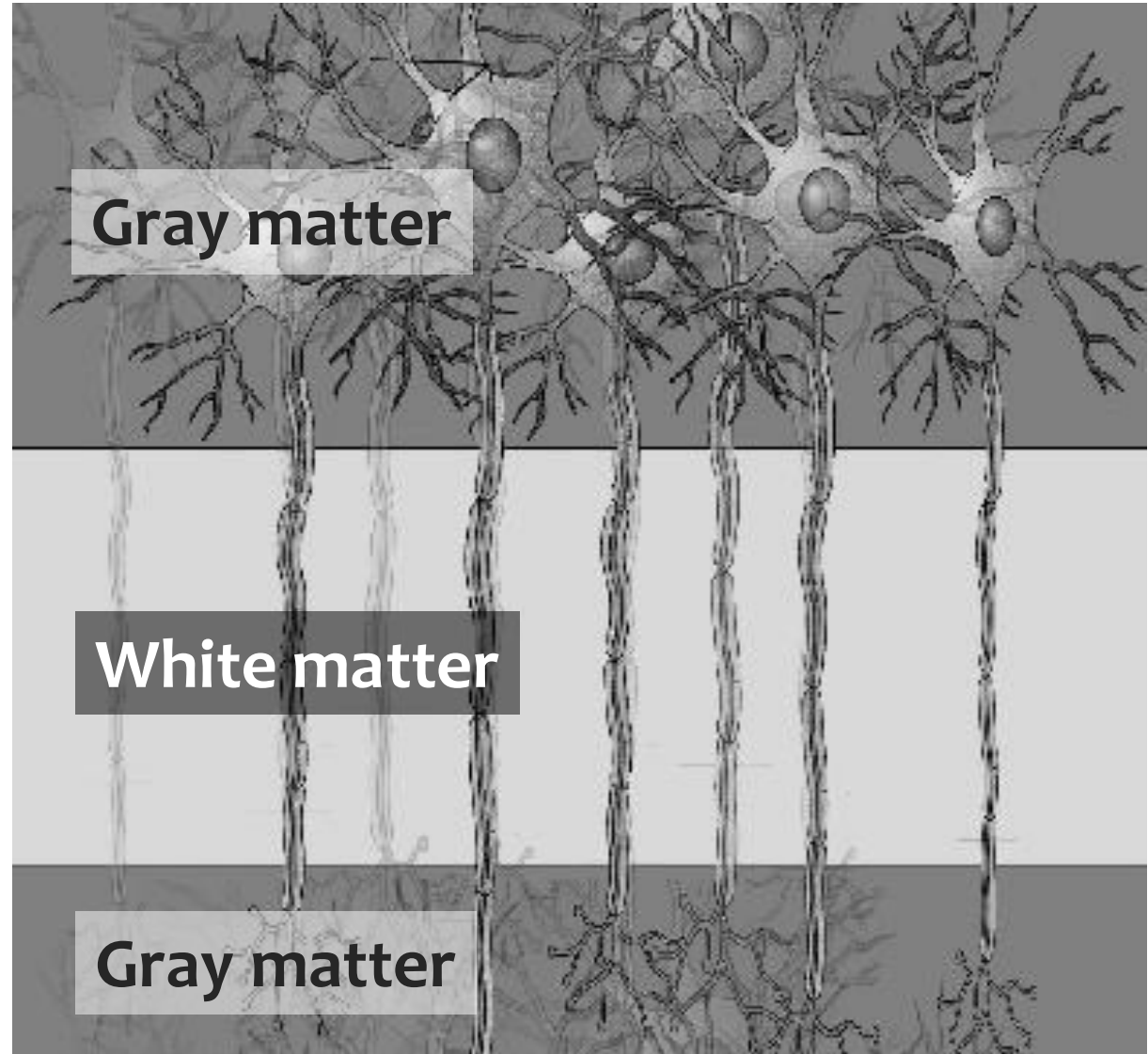
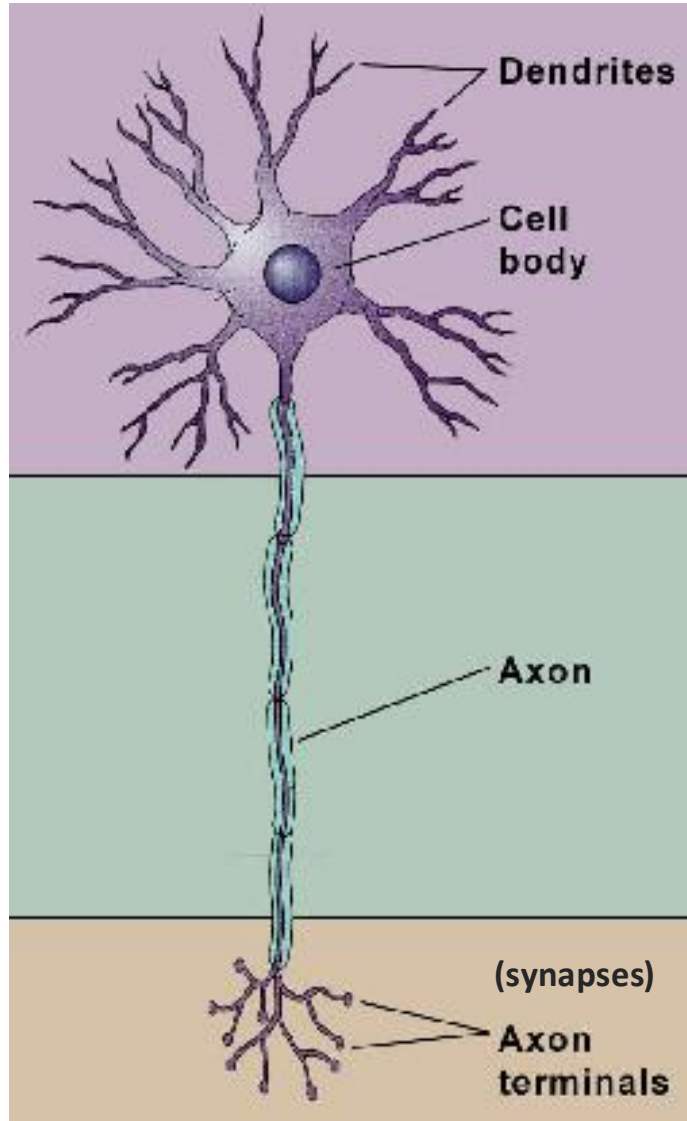


Gray Matter and White Matter

Input Zone

Conducting
Zone

Output Zone



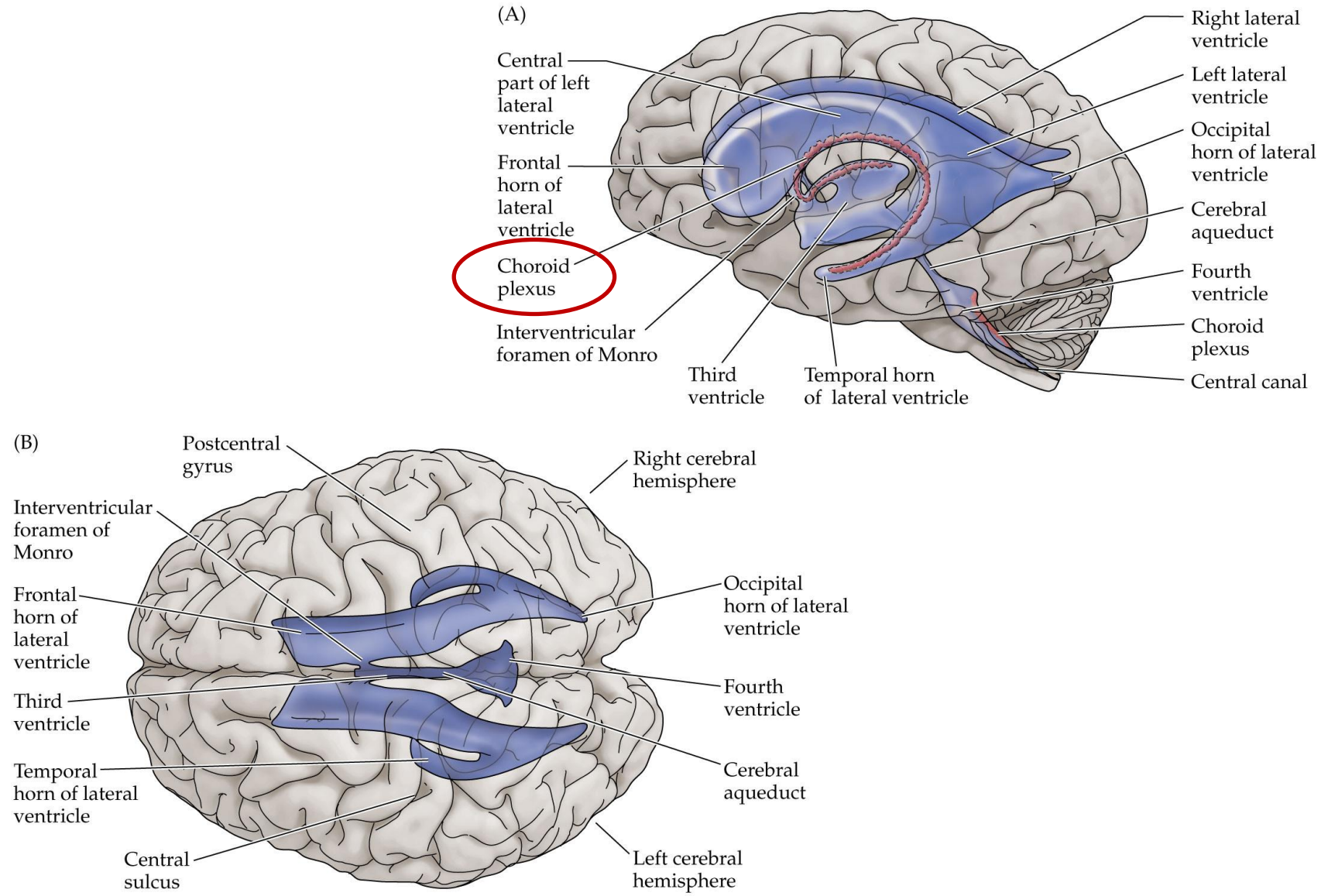
Gray Matter

neurons, their dynamic connections, and
supporting cells



Ventricles

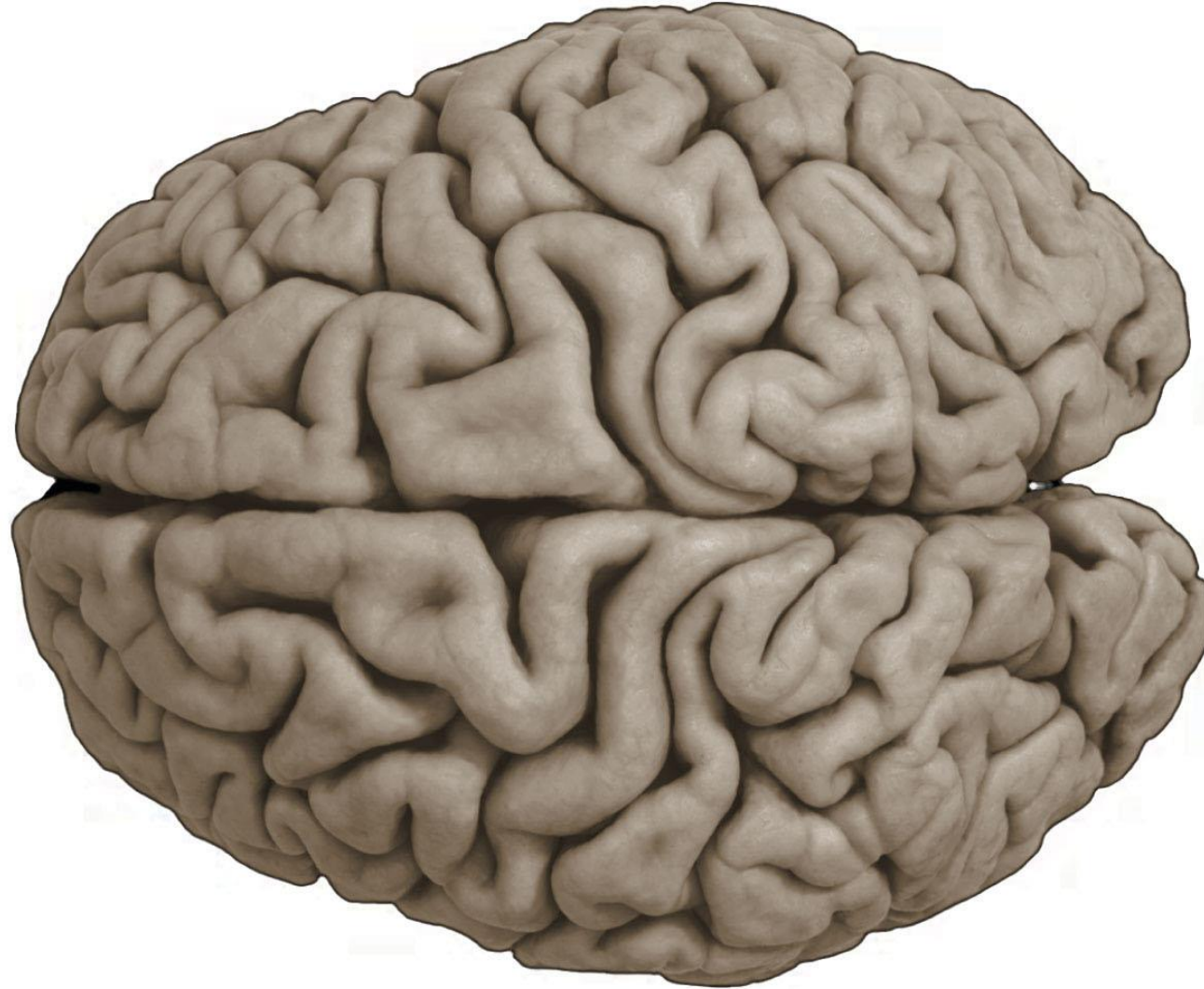
[what became of the lumen of the neural tube]



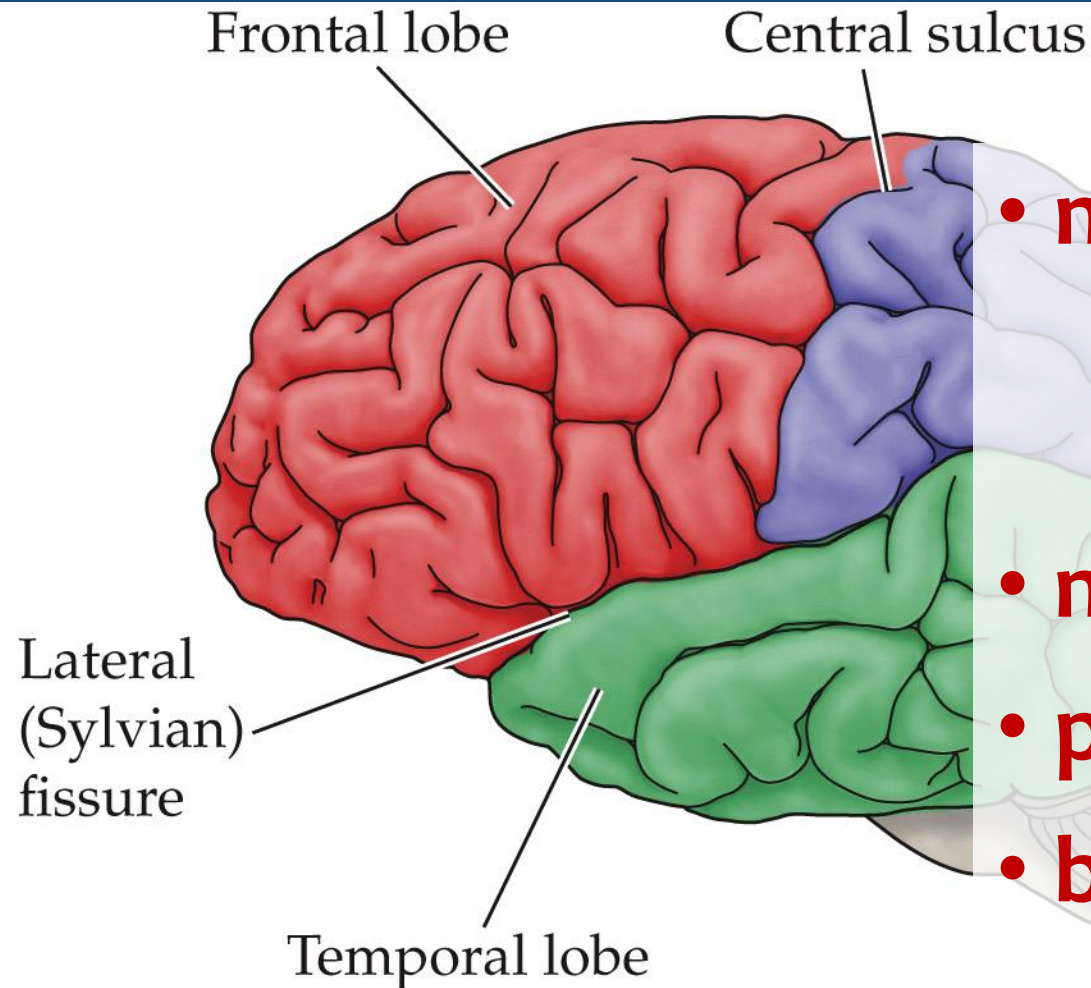
Formation of the human nervous system

Embryonic brain		Adult brain derivatives	Associated ventricular space
Prosencephalon (forebrain)	Telencephalon	Cerebral cortex	Lateral ventricles
		Cerebral nuclei (basal ganglia, amygdala, basal forebrain)	
	Diencephalon	Thalamus	Third ventricle
		Hypothalamus	
		Retina	
Mesencephalon (midbrain)		Superior and inferior colliculi Red nucleus Substantia nigra	Cerebral aqueduct
Rhombencephalon (hindbrain)	Metencephalon	Cerebellum	Fourth ventricle
		Pons	
	Myelencephalon	Medulla oblongata	Fourth ventricle
Spinal cord		Spinal cord	Central canal

Cerebral Hemispheres

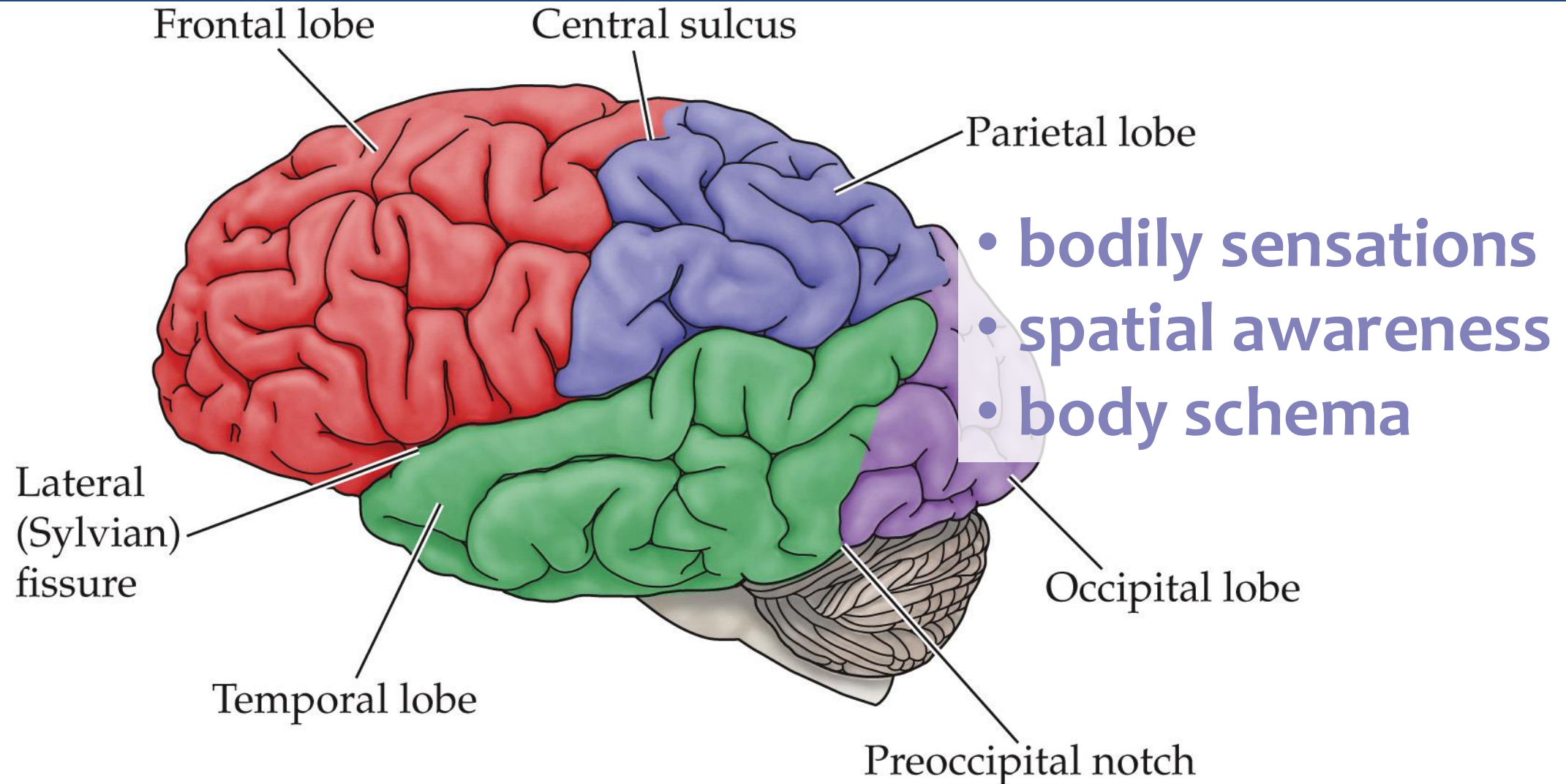


Cerebral Hemispheres

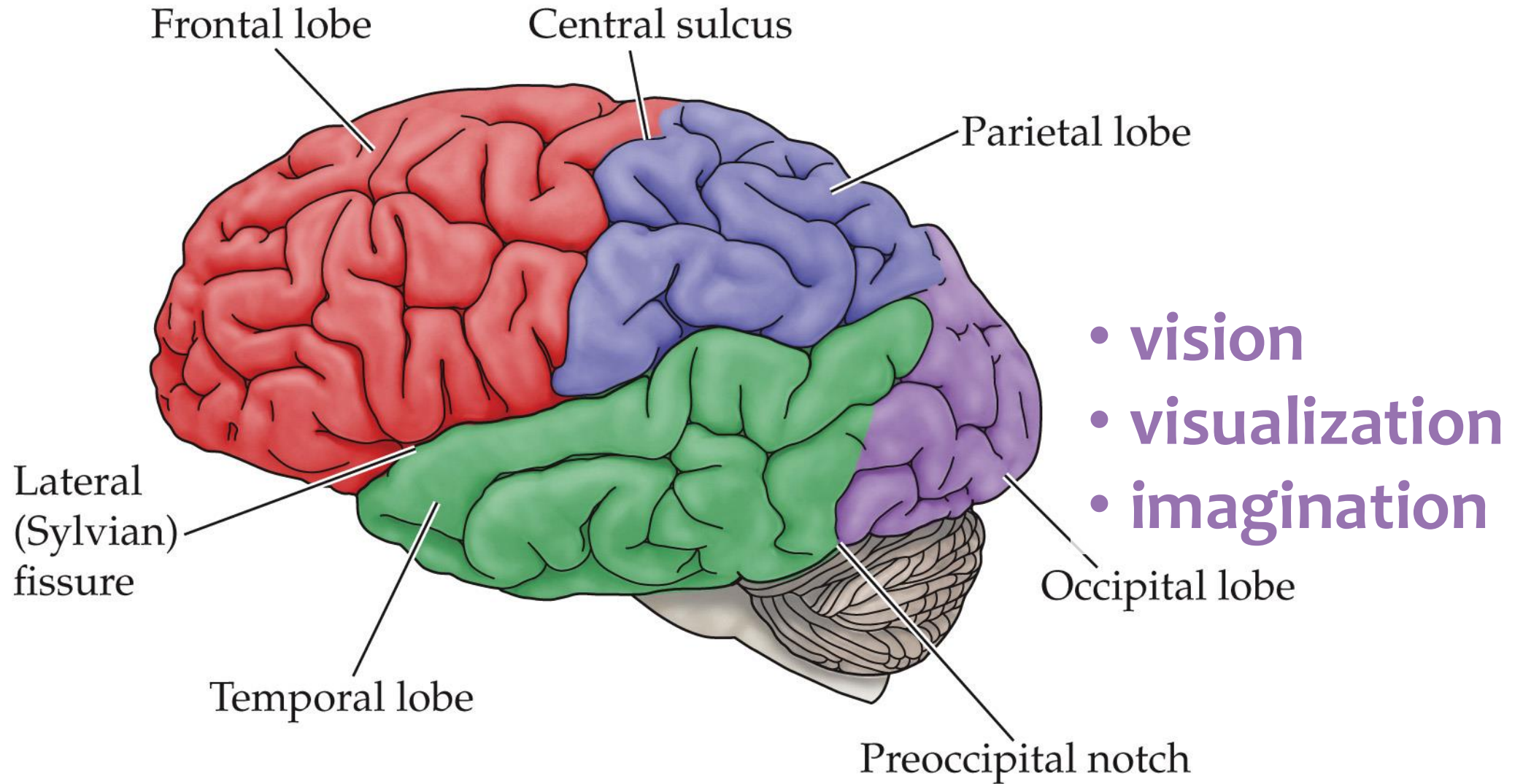


- **making behavior:**
 - movement of mind, body, voice, and emotion
- **making decisions**
- **planning for the future**
- **behavior regulation**
- **social cognition**

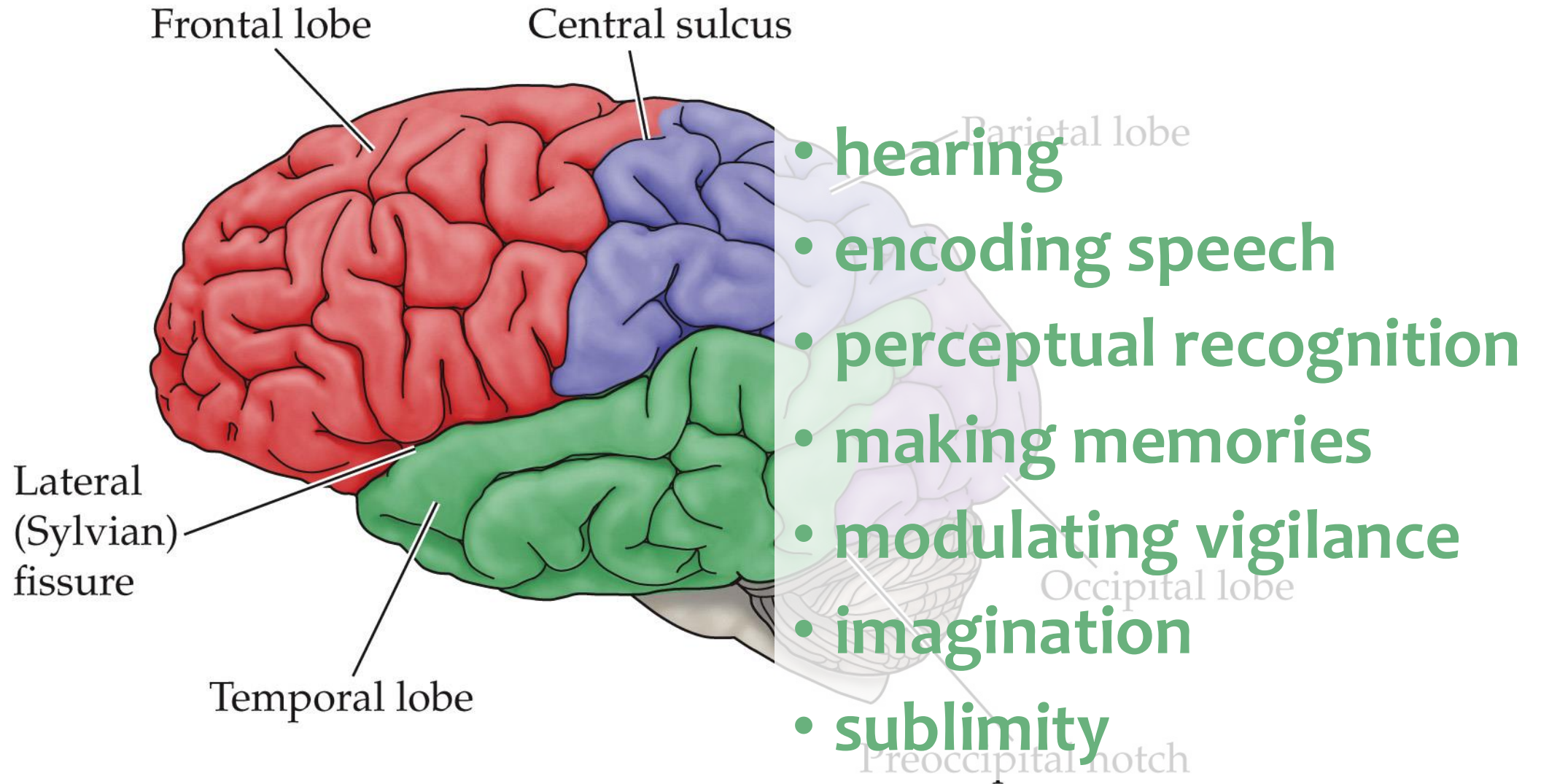
Cerebral Hemispheres



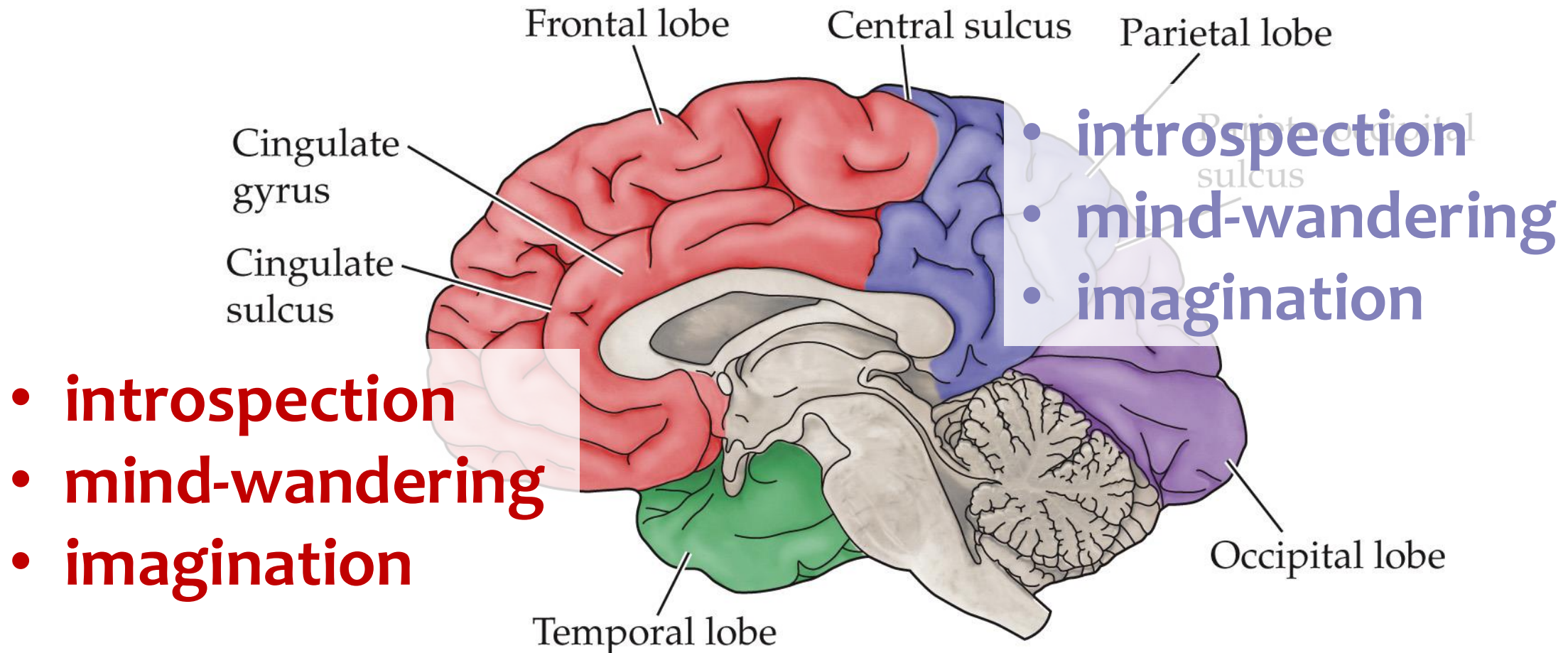
Cerebral Hemispheres



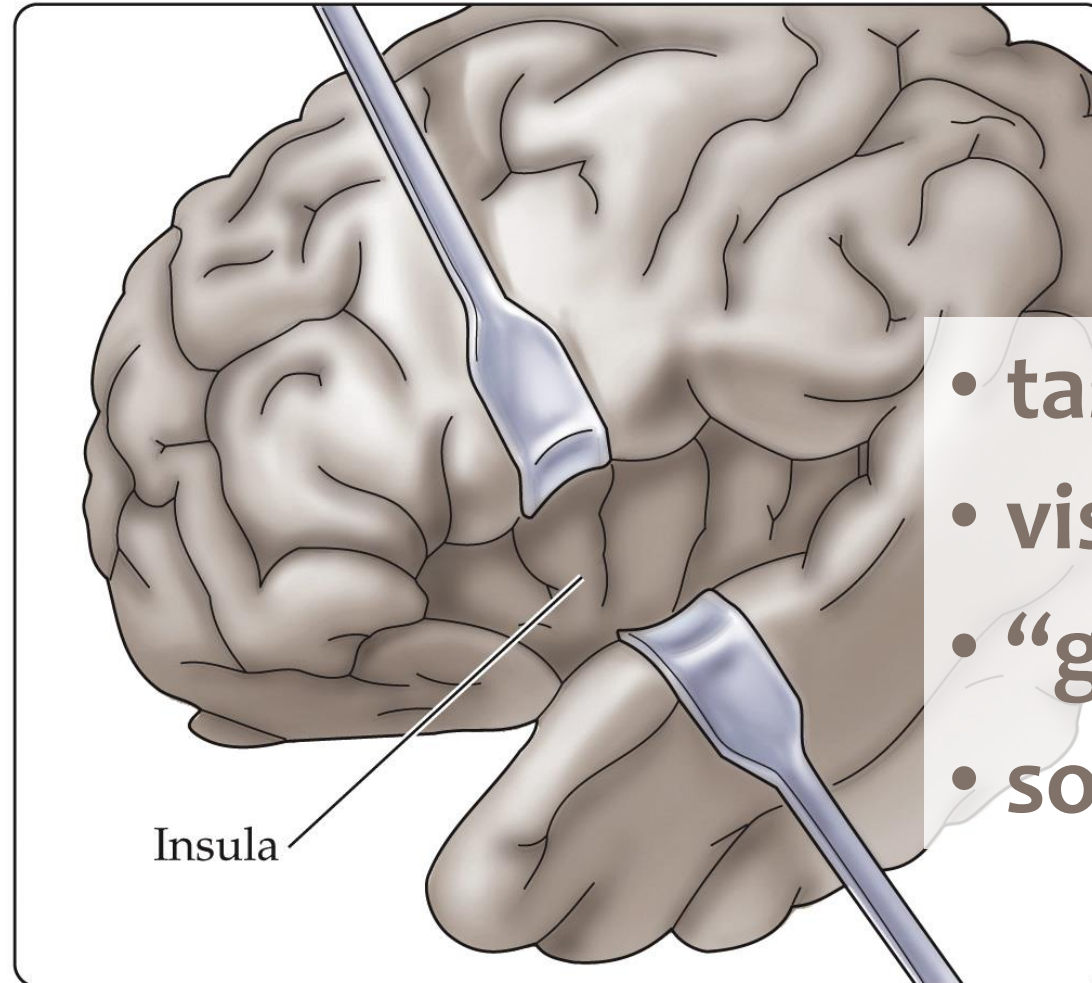
Cerebral Hemispheres



Cerebral Hemispheres

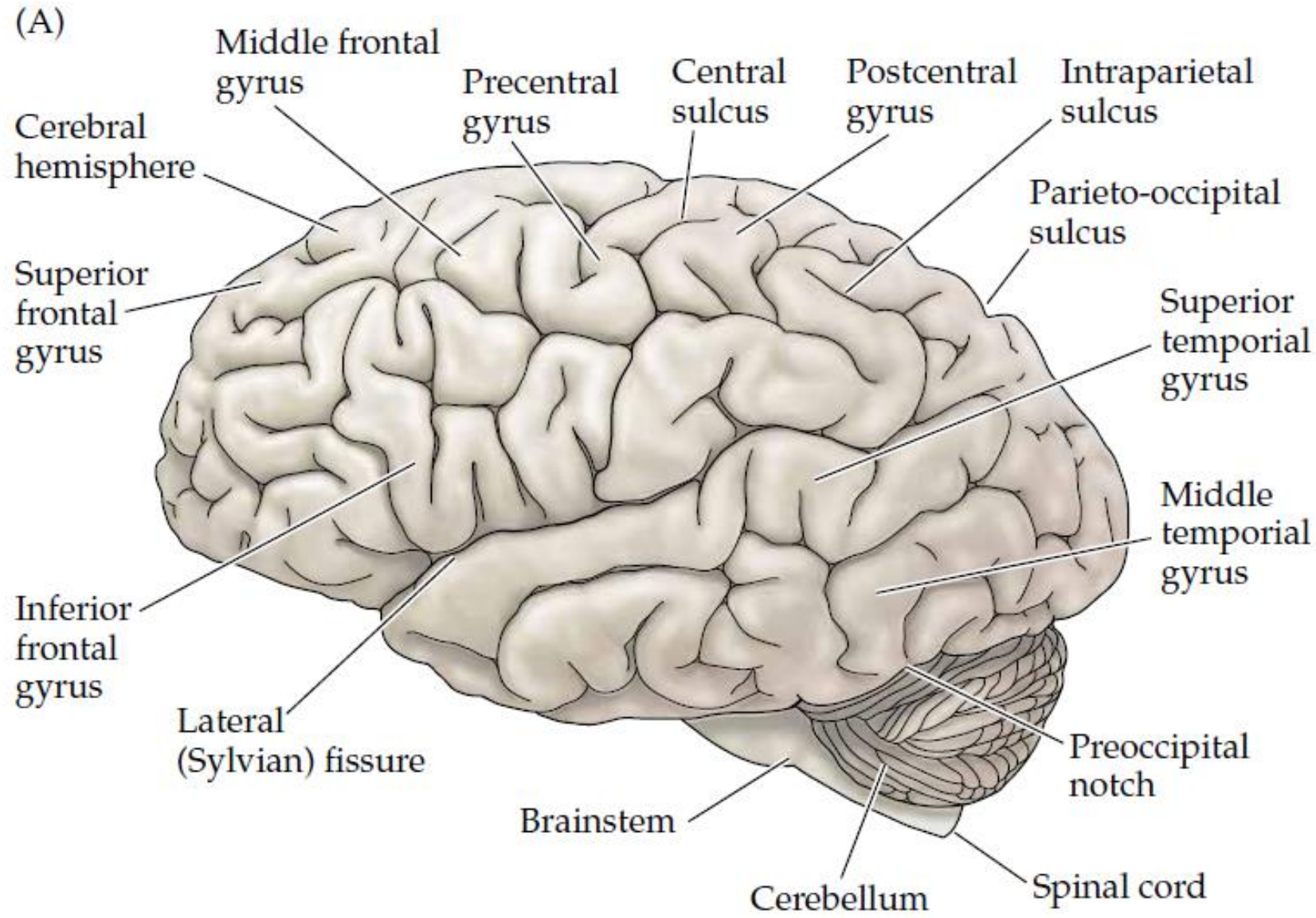


Cerebral Hemispheres

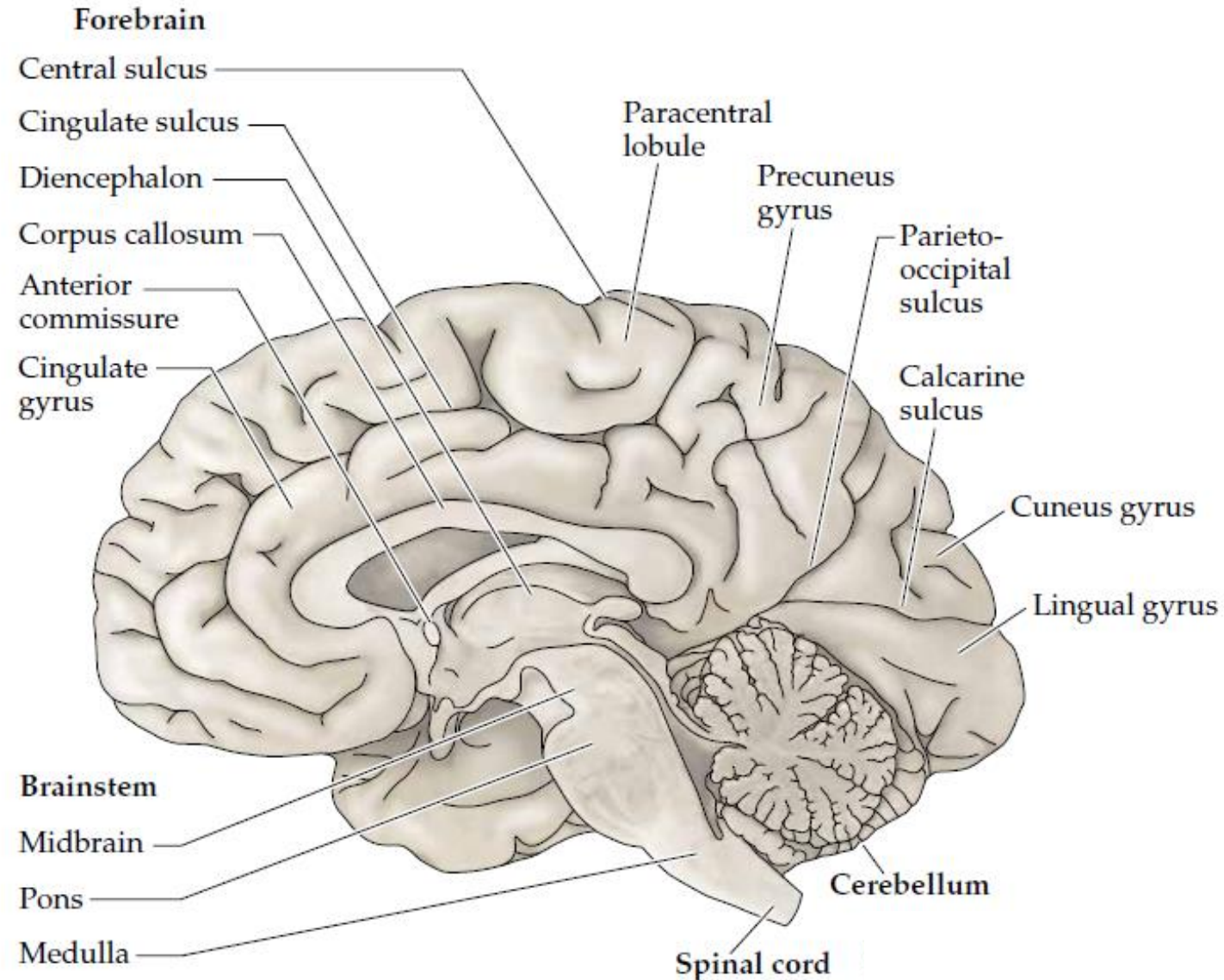


- taste
- visceral sensations
- “gut feelings”
- social cognition

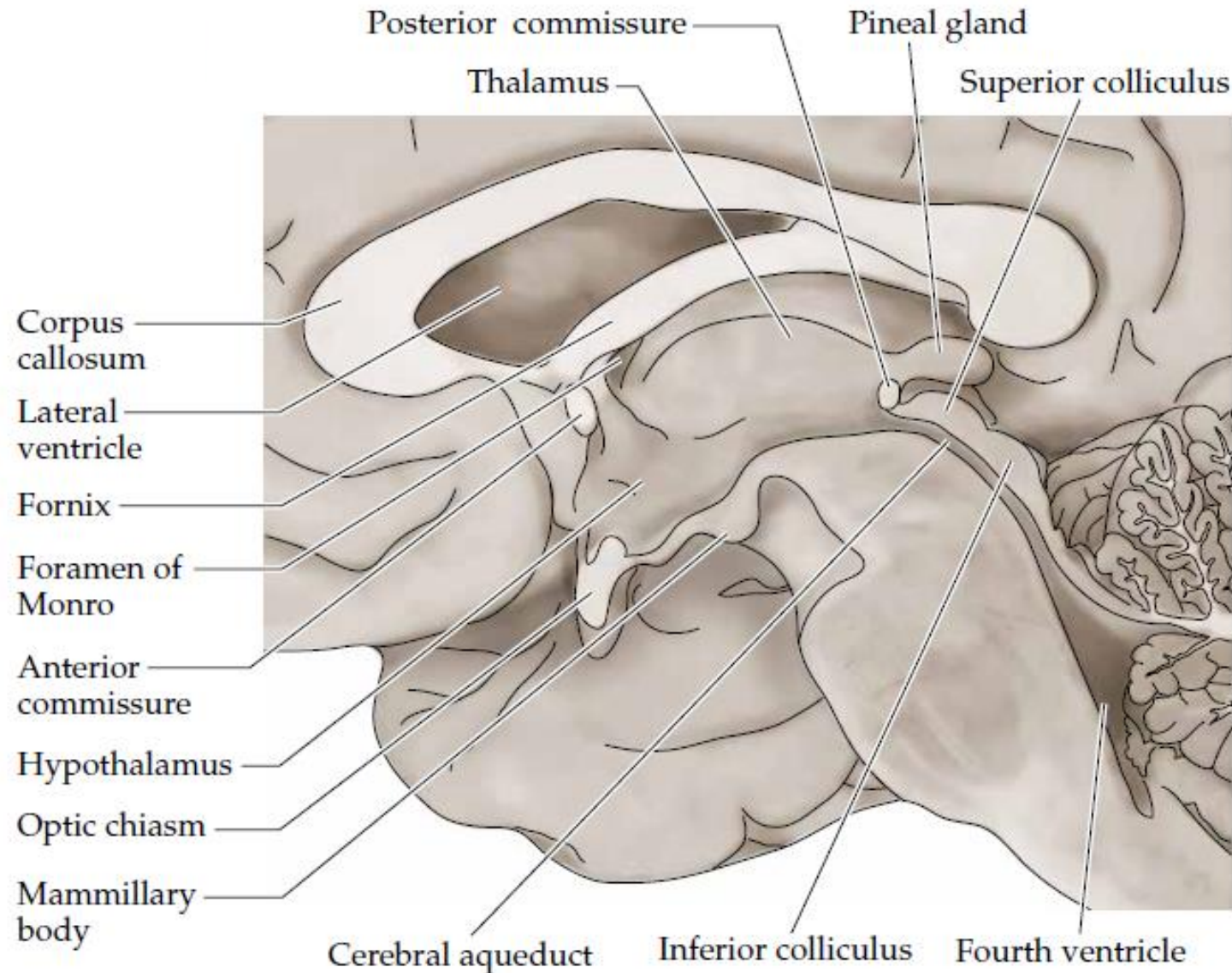
Cerebral Hemispheres



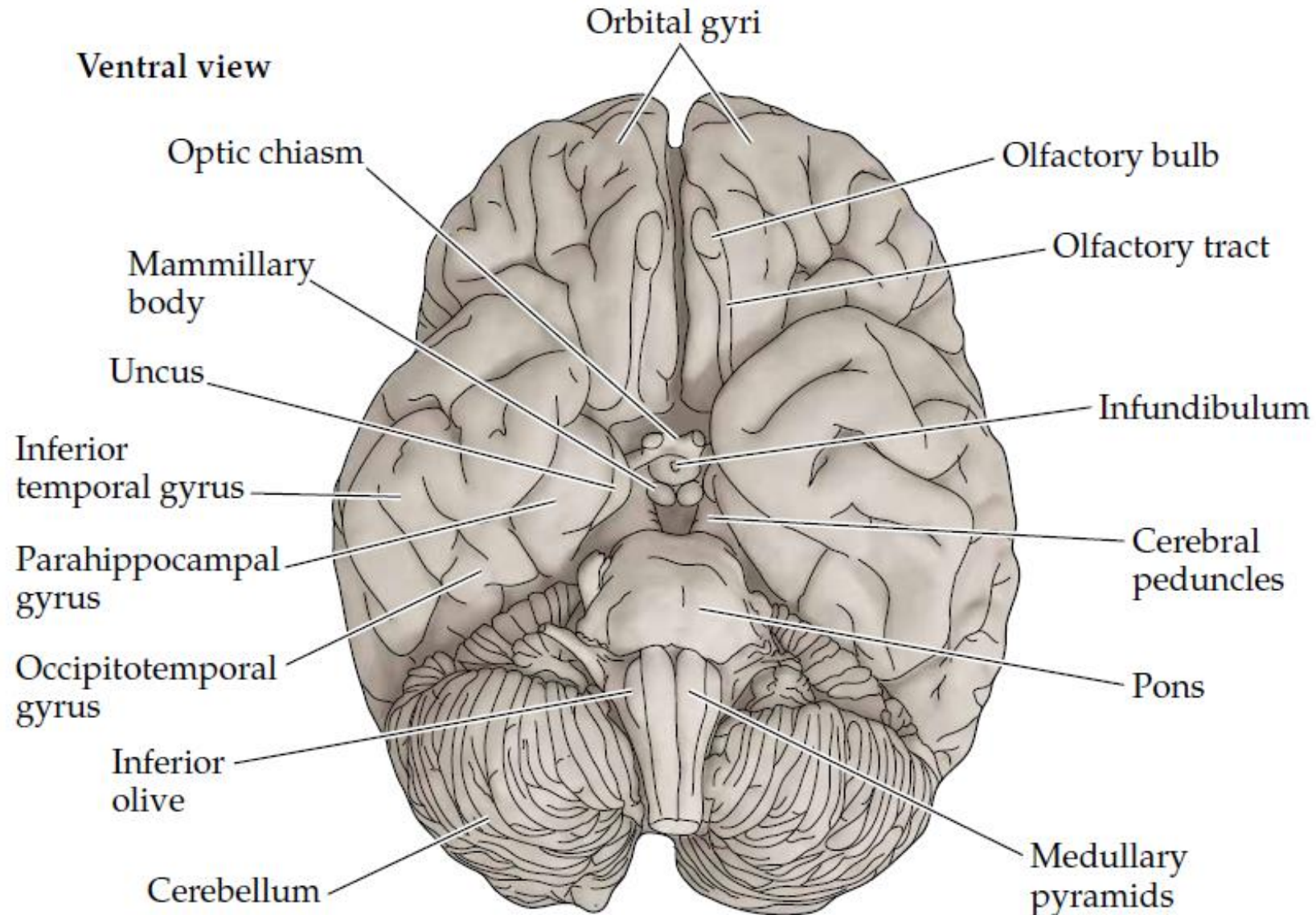
Cerebral Hemispheres



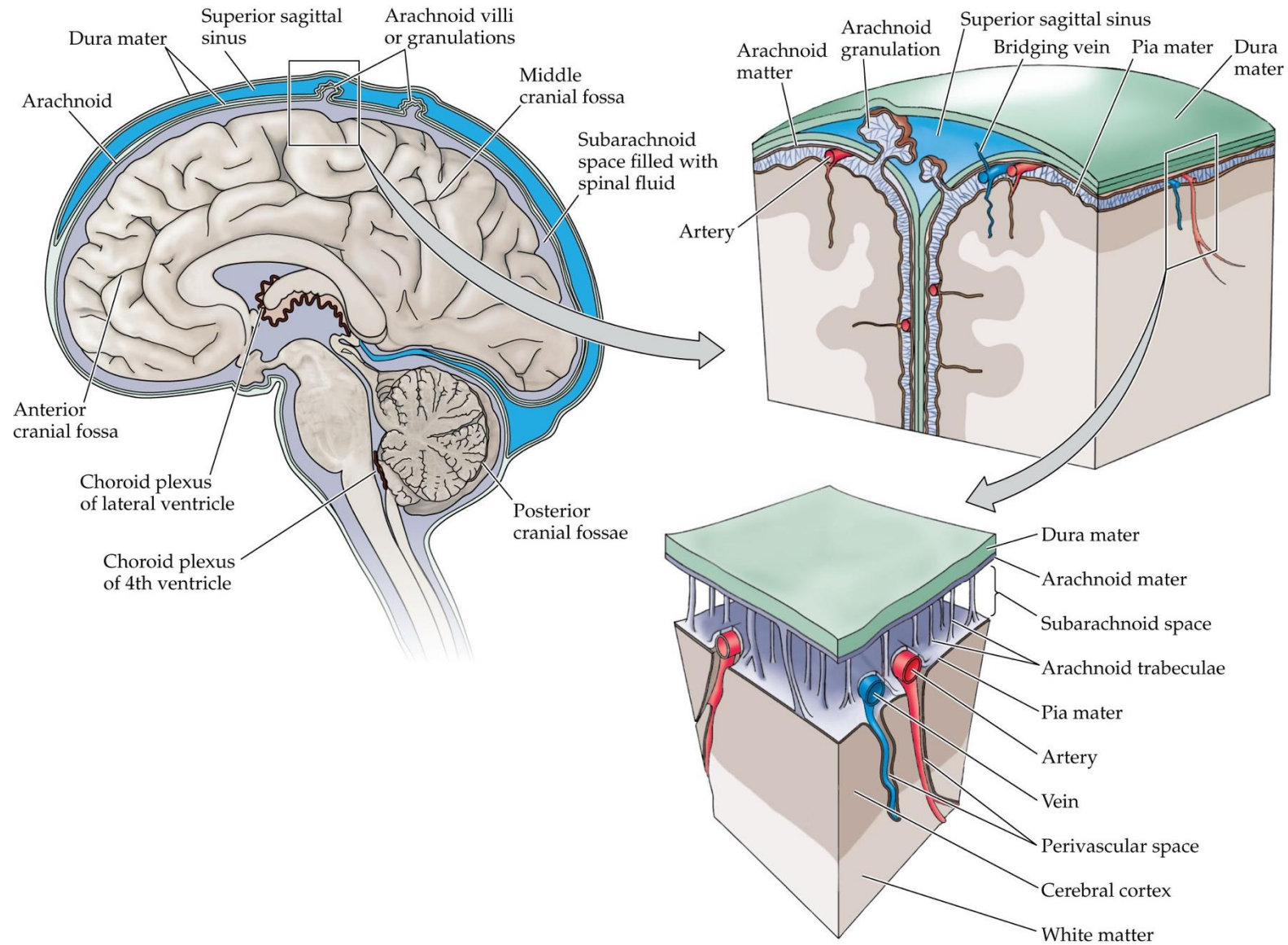
Subcortical Anatomy



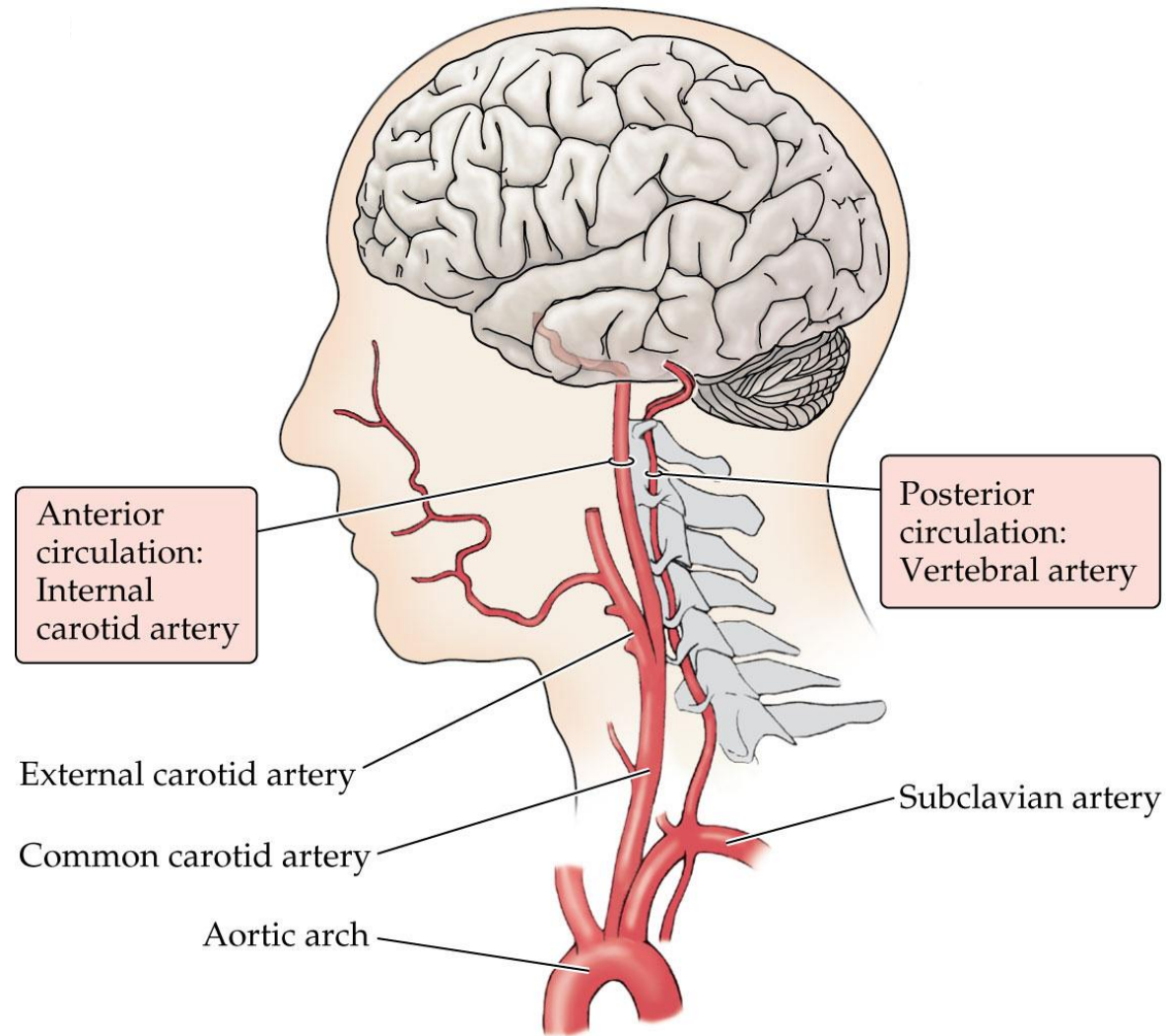
Cerebral Hemispheres



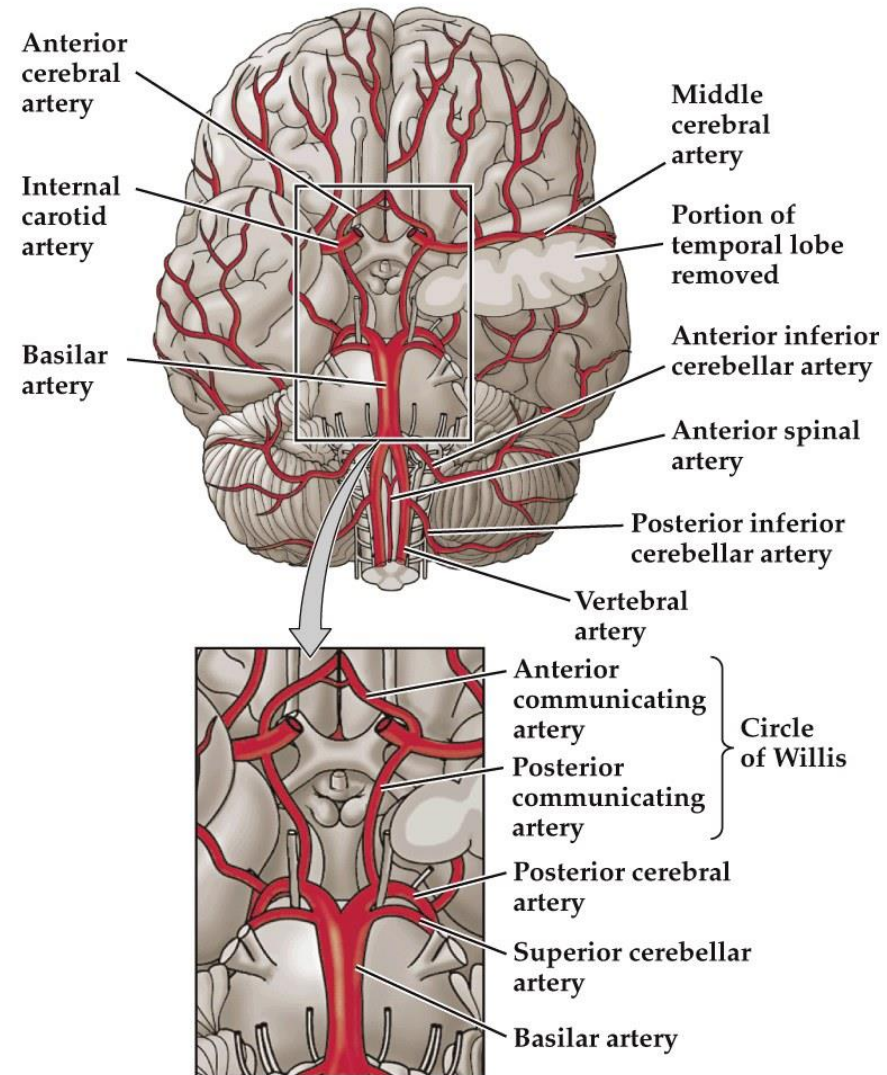
Meninges



Arterial Supply

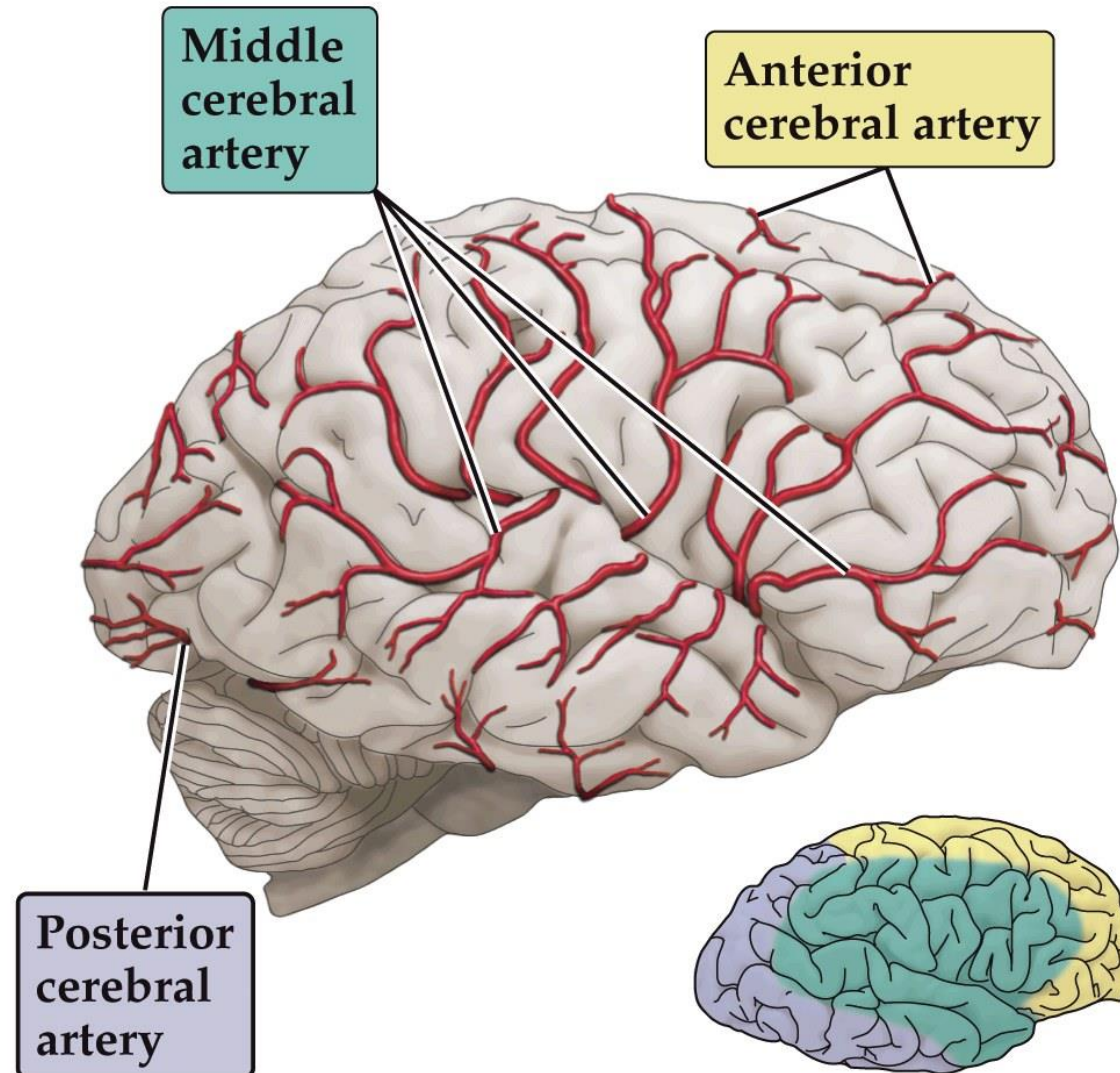


Arterial Supply

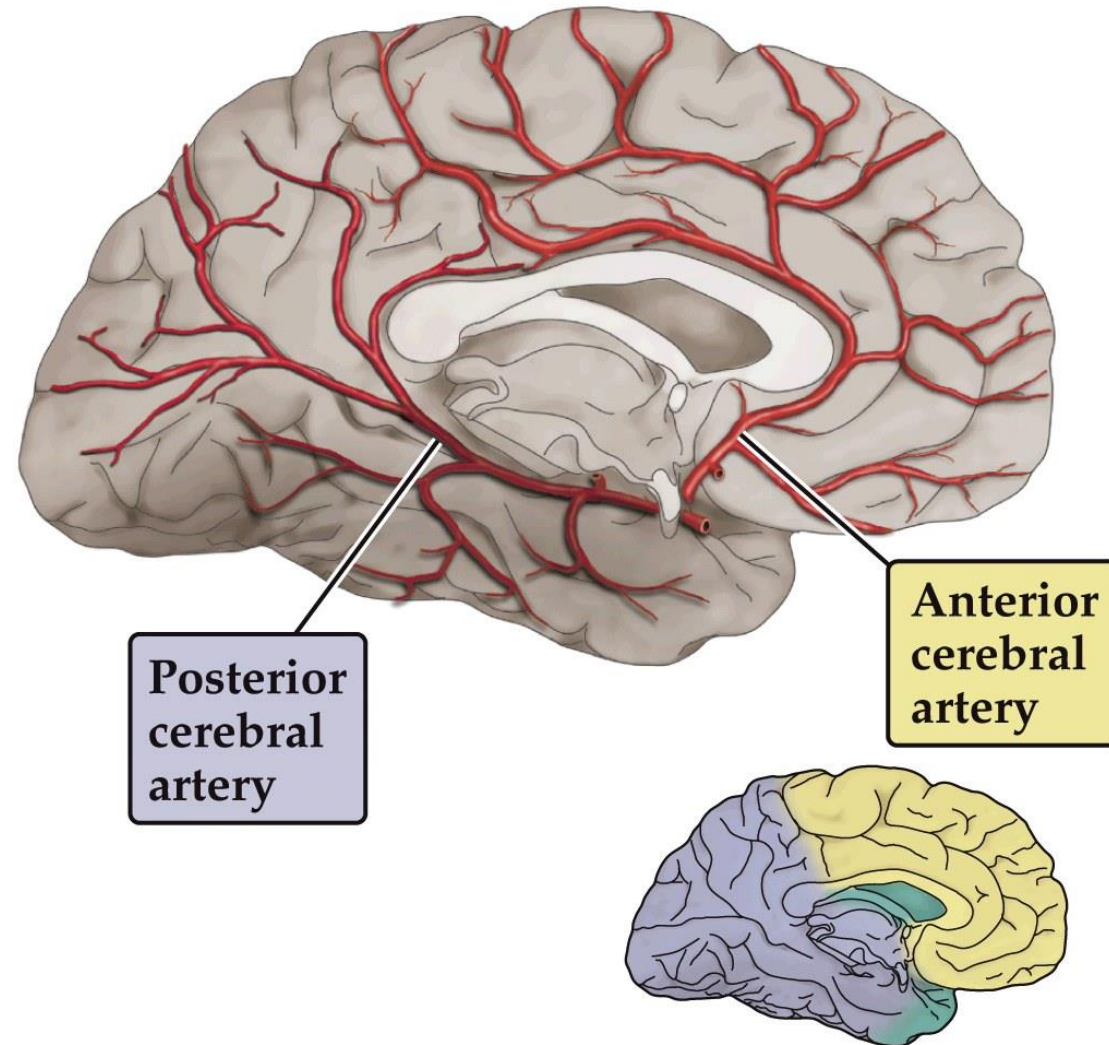


NEUROSCIENCE 5e, Figure A17 (Part 1)

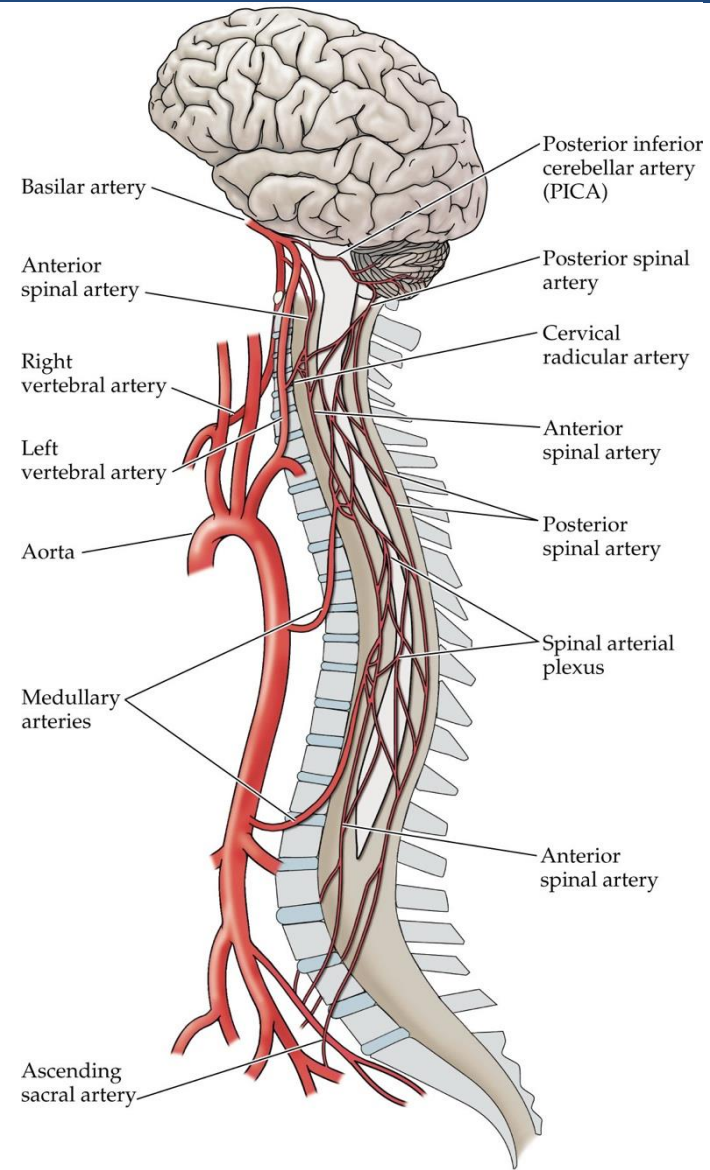
Arterial Supply



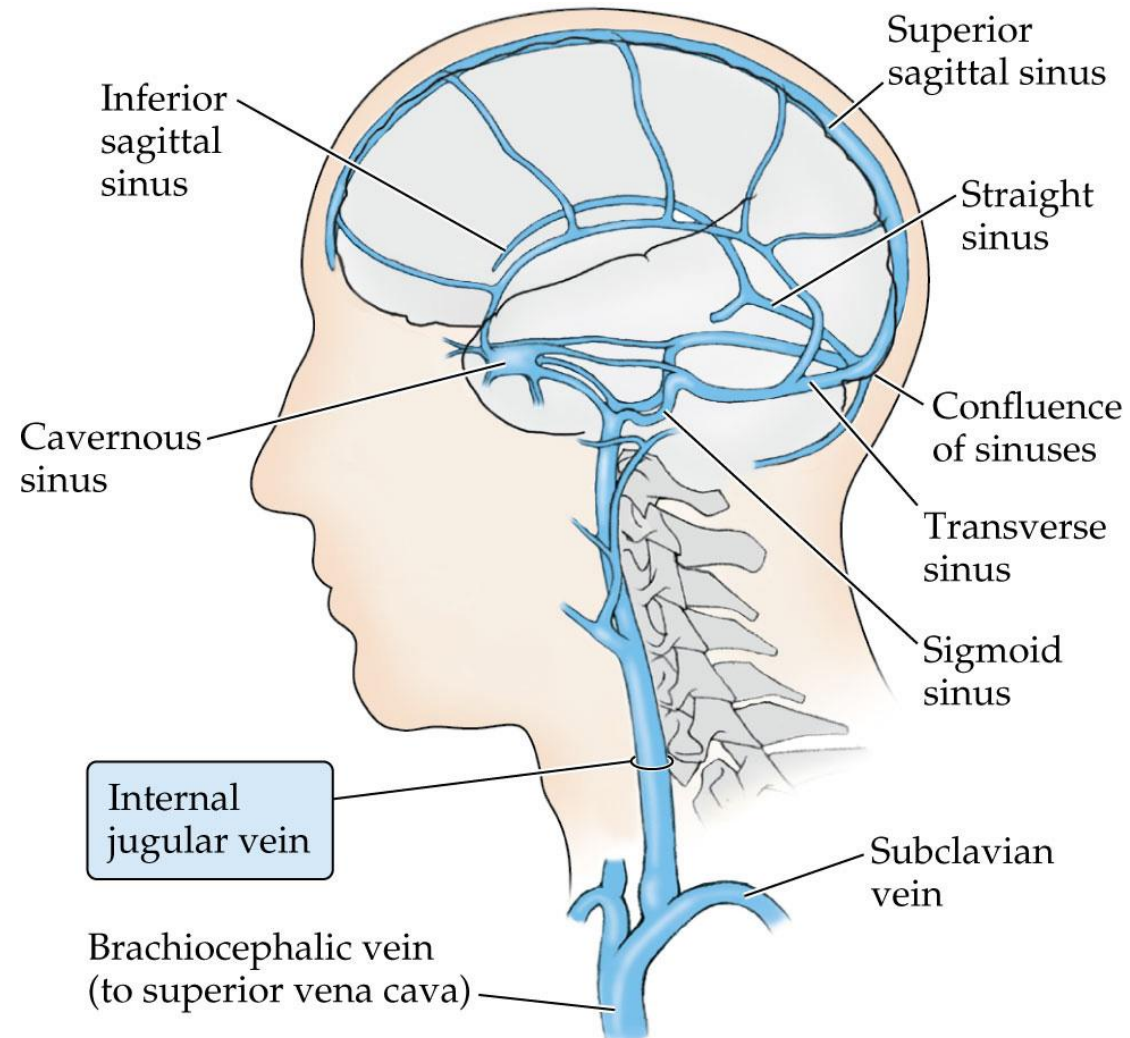
Arterial Supply



Arterial Supply



Venous Drainage



Axes and Directions

